

Cortical Columns, Capsules, and Transformers:

A Typed Benchmark for Part-Whole Representation, Reference Frames, Routing, and Receiver-Relative Dynamics

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August 13, 2026

Preprint; not peer reviewed.

Abstract

A person can recognize a cup while turning it, recover its identity when a handle is hidden, and adjust a grasp as the cup moves relative to the hand. Artificial systems face the same underlying problem: preserving relations among parts, wholes, pose, context, time, and action while their immediate inputs change. Cortical-column theories, Hawkins-style reference-frame models, capsule networks, transformer attention, and Self Aware Networks (SAN) propose materially different operations for that problem, yet they are often compared through metaphors that do not declare units, states, coordinate relations, routing rules, costs, or causal discriminators.

This paper introduces a typed architecture contract and an eight-task Structure-Preserving Routing Benchmark. It first preserves the complete specification, mathematical identifiability analysis, executable generator, compact controls, and mechanism-bearing fixed-encoder calibration. It then reports a frozen trainable application containing seven small models: flat and permutation-equivariant controls, plus recurrent, sensation-location, capsule-routing, transformer-attention, and SAN receiver-state/PWD families. All models receive the same sixteen input scalars, task interface, data partitions, optimizer, epoch count, and frozen seeds; parameter counts span 729 to 3,873, within the declared sixfold band.

The complete support rule required accuracy of at least 0.700, margins of at least 0.030 over both controls, targeted-ablation loss of at least 0.050, and no wrong-frame or nuisance gain above 0.050. None of six target-operation rows passed all four gates. Capsule routing on static assembly (R3/T1) and SAN receiver-state processing on translated perturbation (R5/T8) were favorable but incomplete. Other rows were adverse, including improvement of the sensation-location model under a wrong reference frame and improvement of the attention model after uniform-attention or residual-removal interventions. The application retains 280 model-task-seed cells, 11,200 baseline predictions, 115 intervention cells, 4,600 intervention predictions, 35 checkpoints, 35 traces, 57 new adverse/null rows, and byte-identical copies of the inherited 24- and 38-row negative ledgers. Thirteen deterministic scientific outputs and all 35 learned tensor-state hashes replayed exactly.

A bounded Lean model checks 25 finite interface, provenance, refusal, replay-accounting, and claim-class theorems with no admitted proofs. It proves only consequences of the declared finite encoding; it does not prove a biological premise, source-author parity, population inference, or SAN/PWD validation. The paper therefore supplies an inspectable, falsifiable comparison program and an honestly mixed development result—not a production-model ranking, biological result, or final-test result.

Keywords: cortical columns; Thousand Brains; capsule networks; transformers; attention; reference frames; receiver-relative dynamics; Phase-Wave Differential; Self Aware Networks; architecture benchmark

Reader contract: the vocabulary must be earned

The paper begins with an ordinary recognition problem, identifies the unexplained operation, describes the established architectures in their own terms, and states SAN’s candidate operation in plain language before naming it. A coined term is useful only if a reader can remove it and still say what receives information, what changes, what is transmitted next, and what intervention would distinguish the proposal from a relabeling. Phase-Wave Differential therefore summarizes an already-defined receiver-relative operation; it does not replace phase, rate, power, morphology, reference frame, routing, or downstream consequence.

1 The ordinary problem

Imagine reaching for a cup. The retinal image changes when the eyes move. The handle can disappear behind the cup. The cup occupies one coordinate frame relative to the room and another relative to the hand. Touch supplies evidence that was absent from vision. A successful nervous system does not merely attach the label *cup* to one image. It maintains enough relational structure to predict where the handle should be, revise that prediction after contact, and select an action.

The same difficulty appears in machine learning. A classifier can perform well on familiar images while failing when parts overlap, viewpoints change, an object appears in an unfamiliar composition, or a task requires action-conditioned inference. The scientific problem is therefore not simply pattern recognition. It is **structure-preserving recognition under transformation and intervention**.

Let an episode contain observations ($y_{1:T}$), actions ($a_{1:T}$), latent entities (o), relations ρ , and coordinate transforms ($g_{1:T}$). The target is not only a label but a state that supports prediction and action:

$$\hat{s}_t = f(y_{1:t}, a_{1:t-1}), \quad (\hat{o}_t, \hat{\rho}_t, \hat{g}_t, \hat{y}_{t+1}) = d(\hat{s}_t, a_t). \quad (1)$$

An architecture succeeds when the decoded entity and relations remain useful as the observations change, and when interventions produce the predicted changes rather than merely correlated readouts.

2 What established approaches already provide

2.1 Cortical columns are biological circuits, not tokens

Columnar organization entered neuroscience through physiological and anatomical work showing vertically organized response properties and laminar circuitry [1,2]. Later circuit models proposed recurring cortical motifs [3,4], while modern measurements reveal both conserved laminar regularities and important area-, species-, cell-type-, and task-dependent variation [5–9]. The biologically responsible unit is therefore not a feature token labeled “column.” It is a spatially bounded, experimentally declared population with layers, cell types, dendritic compartments, local and long-range inputs, recurrent state, and outputs.

Laminar traffic is especially relevant. Feedforward and feedback influences can occupy different layers and frequency channels [7,8]. Layer-resolved recordings show that representational information is not distributed identically across laminar compartments [6], and cell-type-specific circuit models can reproduce some population statistics while differing in single-cell detail [9]. These results motivate a column-scale comparison, but they do not make every cortical area one universal microprocessor.

2.2 Hawkins-style columns add object-relative location and lateral constraint

Hawkins and colleagues model active dendritic segments as sparse-pattern recognizers [10]. Their 2017 model joins changing sensation with a location signal inside column-like modules and uses lateral evidence to accelerate object inference [11]. The 2019 framework proposes grid-cell-like location codes in neocortical columns, allowing multiple local models to represent object-relative locations and reach distributed agreement [12]. These are explicit computational hypotheses with activation and learning rules. They are more specific than the general claim that cortex predicts.

The relevant operations are:

1. sparse structural matches on dendritic segments;
2. a location or reference-frame state;
3. temporal integration across sensor movement;
4. lateral support among modules; and
5. learned object models assembled from sensation-location pairs.

2.3 Capsules preserve pose and route by agreement

In capsule networks, a capsule’s vector or matrix represents the presence and instantiation parameters of an entity or part. Lower-level capsules transform their poses into votes for higher-level poses. Iterative agreement changes assignment weights so compatible votes are routed together [13,14]. This makes part-whole relation and viewpoint transformation architectural variables rather than properties a generic classifier must rediscover implicitly.

Capsules therefore supply a clean engineering comparator for relation-sensitive routing. Their internal vectors are mathematical states in a trained model; they are not biological columns simply because both can be described as modules that combine evidence.

2.4 Transformers route content through learned attention

Transformers compute content-dependent weighted mixtures without requiring convolution or recurrence in the original architecture [15]. Cross-attention can compress very large multimodal inputs into a latent bottleneck [16]. Slot Attention adds iterative competition that produces exchangeable, object-centered slots and supports generalization to unseen compositions in its reported tasks [17]. Equivariant networks separately show how known transformation groups can be built into layer-wise structure preservation [18,19].

These systems establish that attention, object binding, pose, and equivariance can be engineered in several ways. Attention weights are not automatically synapses, columns, causal influence, explanation, or conscious access. The comparison must specify which operation is being translated.

3 The SAN source genealogy that motivates the comparison

This paper develops a question that accumulated in stages rather than appearing as one late analogy.

In 2011, Micah Blumberg described the self and experienced world as prediction patterns distributed across the neocortex. In 2012, public Neo Mind Cycle dialogues developed recursive EEG-mediated feedback, multisensory coordination, a predictive “projector” account, and the idea that learned cellular topology constrains what a neuron receives and transmits. These are ancestry stages for distributed predictive representation; they do not contain the later four-architecture benchmark.

On April 11, 2017, the first Neural Lace Podcast publicly connected spatial computation, object construction, prediction, recurrent traffic, and a time-varying three-dimensional neural matrix [35]. Episode 3, published on April 22, developed biological and engineering requirements of a neural interface through discussions of neuroplastic integration, ATP and metabolic demand, artificial-neuron interfacing, and coincidence detection; the bounded local locators are lines 59–113, 283–291, 989–995, and 1219–1225 [20,21]. The exact Hawkins source is the next episode, not Episode 3: on May 5, 2017, Neural Lace Podcast #4 names Jeff Hawkins, *On Intelligence*, Numenta, Hierarchical Temporal Memory, a neocortical hierarchy, and progressive construction from lines and edges to object concepts [31]. The local repaired transcript fixes that passage at lines 31–49; the public YouTube route fixes the episode identity and date, while exact audio timestamps remain outside the admitted quotation set until recovered.

On May 11, 2019, the public Neural Lace Podcast Season 2 Episode 4 article and recording returned to Hawkins-style cortical columns in a more spatial form [22]. The preserved transcript asks whether columns track object orientation, feature relations, and hand-relative change; it then joins predictive feedback, dendritic analog computation, brain-computer-interface difficulty, and three-dimensional semantic segmentation. The relevant transcript spans lines 551–625 and 629–715. Because the source is multi-speaker, this paper paraphrases the bounded operation and does not assign an unattributed sentence to Micah Blumberg.

The public SAN Git history then fixes two additional stages. On June 10, 2022, `a0079z.md` describes arrays whose members detect phase differences, learned morphology that tunes what a cell can receive, and array-to-array re-expression across cortical columns and thalamic bridges. On June 8, 2022, the earliest public Git version of `b0232y` preserves a source conversation in which a dendrite recognizes a temporal pattern, transforms its electrical state, and changes subsequent cellular output; later 2022 revisions connect this with plateau potentials, feedback, and column-scale hypotheses.

SAOv9, fixed in the current repository with SHA-256 `419F4B3C...E7828`, makes the comparison agenda explicit at authored lines 934–944: cortical columns representing patterns; cortico-cortical-thalamic dynamics; PWD effects; artificial-network benefit; “Hinton’s Capsules, Hawkins Columns, Transformer Nets”; and live mapping. SAOv10 repeats that agenda and expands it toward neural architecture history, Thousand Brains, Perceiver-like systems, three-dimensional networks, neural drift, and artificial neurology. The present paper earns its terminology by converting that agenda into a typed, falsifiable benchmark.

4 A typed architecture contract

4.1 Architecture as an open dynamical system

For comparison, define an architecture as

$$\mathcal{A} = \langle U, E, X, M, C, R, L, O, \mathcal{I} \rangle, \quad (2)$$

where (U) is the set of units, (E) the structural and functional edges, (X) fast state, (M) learned or persistent

state, (C) coordinate or relation state, (R) routing, (L) learning, (O) outputs, and $\mathcal{J}$ the permitted interventions. Its execution is

$$(X_{t+1}, M_{t+1}, C_{t+1}, O_t) = F_{\mathcal{A}}(X_t, M_t, C_t, Y_t, A_t; R, L). \quad (3)$$

Every symbol must be bound to measurable variables. In a transformer, a unit might be a token position and its vector. In a capsule model, it is a capsule and pose state. In a Hawkins-style model, it is a modeled cell, mini-column, dendritic segment, or column module. In a biological experiment, it must be a declared cell, compartment, layer-resolved population, or anatomically bounded ensemble.

4.2 Represented relations

For an entity ($\mathbf{o}_i$), define the represented relational state

$$c_i = (p_i, q_i, h_i, \rho_{i1}, \dots, \rho_{in}), \quad (4)$$

where ($\mathbf{p}_i$) is position, ($\mathbf{q}_i$) orientation or pose, ($\mathbf{h}_i$) the selected reference frame, and ρ_{ij} declared part-whole, adjacency, temporal, or action relations. A claim of “reference-frame representation” must identify which components can be decoded, transformed, and causally used.

4.3 Structure-preserving translations

Let Φ map states from architecture $\mathcal{A}$ to architecture $\mathcal{B}$. For operation (k), the translation is accepted to tolerance ε_k only if the mapped execution approximately commutes:

$$d_k(\Phi_k(F_{\mathcal{A}}(s, a)), F_{\mathcal{B}}(\Phi_k(s), \phi_k(a))) \leq \varepsilon_k. \quad (5)$$

Ordinary input-output agreement is insufficient. The intervention must also translate:

$$d_k\left(\Phi_k(F_{\mathcal{A}}^{\text{do}(I)}(s, a)), F_{\mathcal{B}}^{\text{do}(\phi_I(I))}(\Phi_k(s), \phi_k(a))\right) \leq \varepsilon_k^I. \quad (6)$$

This is the anti-metaphor gate. Two systems can give the same answer for different reasons. A biological column and a capsule are equivalent only for the bounded operations that satisfy both execution and intervention tests.

4.4 Residual vector and graded equivalence

For each architecture, score a declared operation vector

$$v_{\mathcal{A}} = [u, e, c, x, m, r, f, \ell, b, d, i], \quad (7)$$

corresponding to unit, edge, coordinate, fast state, memory, routing, feedback, learning, embodiment, drift, and intervention. The weighted operation-preservation score is

$$S(\mathcal{A}, \mathcal{B}) = \frac{\sum_k w_k P_k(\mathcal{A}, \mathcal{B})}{\sum_k w_k}, \quad P_k \in [0, 1]. \quad (8)$$

The residual $(1 - P_k)$ is scientifically useful. It identifies what an experiment must distinguish. It is not a demotion of either architecture and does not require a whole-system verdict.

5 Five architecture families under the same contract

5.1 Biological cortical circuit

Let $(x_{l,c}(t))$ denote measured state in layer (l) and cell class (c). A minimal open-circuit abstraction is

$$x_{l,c}(t+1) = \sigma_{l,c} \left(\sum_{l',c'} W_{l,c,l',c'} x_{l',c'}(t) + B_{l,c} y_t + T_{l,c} z_t + \eta_{l,c}(t) \right), \quad (9)$$

where (B) carries local/feedforward input, (T) contextual, corticocortical, or thalamic input, and (W) recurrent intracortical interaction. The meaning of a “column” is fixed by the probe geometry, anatomy, receptive-field mapping, and layer/cell-type assignment. The model remains incomplete until dendritic and glial variables relevant to the experiment are declared.

5.2 Hawkins-style structural match and lateral consensus

For segment (b) of modeled cell (i) under hypothesis (h), let

$$m_{ibh}(t) = \mathbf{1}[\langle D_{ibh}, s(t) \rangle \geq \theta_{ibh}] \quad (10)$$

represent a sparse structural match. A module updates an object hypothesis from sensation (s_t), location ℓ_t , prior state, and lateral evidence:

$$h_j(t+1) = H_j(s_t, \ell_t, h_j(t), \{h_k(t)\}_{k \neq j}). \quad (11)$$

The distinctive test is whether object-relative location and lateral agreement improve rapid inference under movement and partial evidence beyond matched recurrent sparse baselines.

5.3 Capsule votes and routing

For lower capsule (i) and candidate whole (j),

$$\hat{u}_{j|i} = W_{ij} u_i \quad (12)$$

is a transformed vote. Routing assigns coupling coefficients

$$c_{ij} = \text{softmax}_j(b_{ij}), \quad b_{ij} \leftarrow b_{ij} + \hat{u}_{j|i}^\top v_j. \quad (13)$$

The distinctive test is whether agreement among transformed part poses improves overlapping-object, viewpoint, and part-whole recombination tasks beyond matched attention and equivariant baselines.

5.4 Transformer and object-centric attention

Scaled dot-product attention is

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V. \quad (14)$$

A recurrent or iterative attention architecture can be written

$$z^{(r+1)} = G\left(z^{(r)}, \text{Attn}(Q(z^{(r)}), K(y), V(y))\right). \quad (15)$$

The distinctive tests concern learned content routing, latent bottlenecks, exchangeable object slots, and the extent to which coordinate relations emerge or must be supplied through position, pose, or equivariant structure.

5.5 SAN: persistent receiver structure plus fast relational departure

Before introducing SAN vocabulary, consider what a biological receiver must do. A cell or local population is already in a state when new input arrives. Its dendritic morphology, synaptic history, inhibition, membrane state, recent activity, and network context affect the consequence of that input. The same arriving event can therefore have different effects at different receivers or times. A useful signal is not only the sender's activity; it is the measurable difference between what arrives, what the receiver was prepared to receive, and what relation was expected on that route.

Let persistent receiver state be ($M_i(t)$), fast state ($q_i(t)$), context or reference frame ($h_i(t)$), and incoming events ($y_{ji}(t - \tau_{ji})$). The effective drive is

$$d_i(t) = G_i(M_i(t), q_i(t), h_i(t), \{y_{ji}(t - \tau_{ji})\}_{j \in \mathcal{N}_i}). \quad (16)$$

Define an expected receiver-relative relation $\mu_i(t)$ learned from a training partition and a measured event vector $r_i(t)$. Because phase is circular while the other coordinates are linear or categorical, the departure uses a component-specific product-space difference rather than raw vector subtraction:

$$r_i(t) = [\phi_i, \omega_i, \log A_i, w_i, e_i], \quad \delta_i(t) = \Delta_{\mathcal{R}}(r_i(t), \mu_i(t)). \quad (17)$$

Only now is the name useful: SAN calls a causally effective, receiver-relative departure of this kind a **Phase-Wave Differential**. Phase is one declared coordinate; frequency, amplitude or power, waveform/transmission state, sign, delay, and receiver readiness can also matter. A departure has information value relative to a learned conditional distribution,

$$I_i(t) = -\log_2 p(r_i(t) \mid M_i(t), q_i(t), h_i(t), \mathcal{H}_t), \quad (18)$$

where $\Delta_{\mathcal{R}}$ wraps phase on the circle, subtracts compatible scalar coordinates after unit-preserving calibration, and retains sign or category without inventing an ordinal distance. In Equation 18, p is either a preregistered probability mass or a density relative to a frozen reference measure and binning rule; changing that measure after seeing test data is prohibited. Surprisal becomes biologically meaningful only when a tuned receiver changes state or action. The full SAN update is

$$(q'_i, M'_i, h'_i, o_i) = F_i(q_i, M_i, h_i, d_i, \delta_i; \mathcal{N}_i). \quad (19)$$

This two-timescale construction is SAN’s proposed residual: learned structure constrains admissible matches; fast receiver-relative differences select, revise, or propagate an active route; recurrent exchange and action test the resulting representation.

6 Typed crosswalk

Dimension	Biological cortical circuit	Hawkins-style column	Capsule network	Transformer / slots	SAN neural array
Unit	Cell, compartment, layer-resolved ensemble	Modeled cells, mini-columns, column modules	Vector or matrix capsule	Token/latent/slot vector	Cell, dendritic receiver, oscillating group, array
Relation	Empirically decoded and circuit-dependent	Sensation plus object-relative location	Part pose transformed into whole-pose vote	Learned content relation; positional structure supplied or learned	Receiver-relative event plus body/object/task context
Routing	Synaptic, laminar, corticocortical, thalamocortical	Sparse match, prediction, lateral support	Iterative routing by agreement	Soft attention or iterative competitive attention	Connectivity, tuning, inhibition, phase compatibility, recurrence
Persistent state	Morphology, synapses, cell state, circuit plasticity	Learned dendritic segments and object models	Trained weights and capsule parameters	Trained weights; optional recurrent/working state	Morphology, synapses, receptor state, learned ensemble structure
Fast state	Voltage, spikes, bursts, field and transmitter variables	Active/predictive/inactive states	Capsule activation and pose	Token/latent activations and attention	Ongoing state plus PWD residual vector
Embodiment	Intrinsic brain-body loop	Sensor movement central to object learning	Usually task-dependent, often image-centered	Optional; not intrinsic to base transformer	Action and returned sensory/body evidence are constitutive tests
Causal discriminator	Layer/cell/compartment perturbation and rescue	Remove location or lateral consensus	Remove pose transform or routing iterations	Remove attention, position, recurrence, or slot competition	Preserve rate/power while perturbing receiver-relative relation or learned receiver state

The table is not a rank order. It is a test contract. Each row makes clear where a mapping may be strong, partial, or undefined.

7 The Structure-Preserving Routing Benchmark

7.1 Dataset and tasks

Construct a procedurally generated **Moving Assembly World** in which objects contain reusable parts joined by typed relations. Each episode provides RGB-D or point-cloud observations, optional touch/proprioception, camera and effector motion, controlled occlusion, and a ground-truth scene graph. Training and test partitions separate familiar objects, unfamiliar viewpoints, unseen part combinations, and altered action policies.

The benchmark distribution is

$$\mathcal{D} = \{(y_{1:T}^{(m)}, a_{1:T}^{(m)}, o^{(m)}, \rho^{(m)}, g_{1:T}^{(m)})\}_{m=1}^N. \quad (20)$$

Eight tasks are required:

1. static assembly from reusable parts and distractors;
2. pose transfer across held-out translation and rotation, with reflection only where declared valid;
3. part-whole recovery under partial occlusion;
4. part reassignment in which the same local part belongs to a different whole;
5. active inspection under a bounded action budget;
6. context reversal using identical current observations after different histories;
7. drift and re-registration under a stable world relation; and
8. translated perturbation, matched rescue, and mismatched-rescue control.

7.2 Model families

The application implements:

- **M0**: paired feedforward controls: a convolutional or MLP baseline and a parameter-reported equivariant subcondition over a declared transformation group;
- **M1**: recurrent sparse/column-like baseline without explicit reference frames;
- **M2**: Hawkins-style sensation-location and lateral-consensus model;
- **M3**: capsule model with pose votes and routing;
- **M4**: transformer/Perceiver/Slot-Attention family; and
- **M5**: SAN-inspired two-timescale receiver model with structural state, recurrence, and declared residual messages.

No implementation is treated as a complete biological brain. M2 and M5 are operational models of selected hypotheses.

7.3 Fairness contract

Models must be matched across more than parameter count. Let

$$B_m = (P_m, F_m, T_m, R_m, S_m, E_m), \quad (21)$$

where (P) is trainable parameters, (F) training compute, (T) training examples or information exposure, (R) recurrent iterations, (S) mutable state capacity, and (E) embodied interaction opportunities. A comparison is admitted only when every dimension is matched or its difference is reported and tested through a scaling curve.

7.4 Metrics

For transformation (g) with output representation $\rho(g)$, equivariance error is

$$\mathcal{E}_{\text{eq}} = \mathbb{E}_{x,g} [\|f(T_g x) - \rho(g)f(x)\|_2]. \quad (22)$$

Part-whole relation score is

$$\mathcal{S}_\rho = 1 - \frac{1}{|\rho|} \sum_{(i,j) \in \rho} \ell_\rho(\hat{\rho}_{ij}, \rho_{ij}). \quad (23)$$

where $\ell_\rho \in [0, 1]$ is fixed per relation type and $|\rho| > 0$. Episodes without an admissible relation are marked not applicable rather than assigned a perfect score.

Temporal prediction error is

$$\mathcal{E}_{\text{pred}} = \frac{1}{T-1} \sum_{t=1}^{T-1} \ell_y(\hat{y}_{t+1}, y_{t+1}). \quad (24)$$

Causal necessity of component (k) is measured against a matched sham intervention, with (J) preregistered as a higher-is-better task score:

$$\Delta_k = \mathbb{E}[J \mid \text{do}(I_k^{\text{sham}})] - \mathbb{E}[J \mid \text{do}(I_k)]. \quad (25)$$

Representational drift is measured after learning or context change while holding task identity constant:

$$\mathcal{D}_{\text{drift}}(\tau) = 1 - \text{CKA}(Z_t, Z_{t+\tau}), \quad (26)$$

and is reported together with behavior so code change is separated from loss of function. CKA is computed only over matched episode or stimulus identities; unequal sample sets require a preregistered alignment procedure and separate sensitivity report. Longitudinal studies show that representational drift can depend on stimulus and preparation, and that stable task information can remain decodable from a population whose individual tuning changes [32–34]. Those results do not license the assumption that every drift is harmless: a fixed decoder may still degrade, so the benchmark reports single-session, multi-session, aligned, and behavior-linked readouts separately.

Information efficiency is

$$\eta_m = \frac{I(O, \rho; Z_m)}{\tilde{F}_m + \lambda_P \tilde{P}_m + \lambda_S \tilde{S}_m}, \quad (27)$$

where $\tilde{F}_m$, $\tilde{P}_m$, and $\tilde{S}_m$ are nonnegative dimensionless ratios to positive, frozen reference budgets for training compute, parameter count, and mutable-state capacity, and λ_P and λ_S are dimensionless preregistered weights. The score is undefined when the total denominator is zero. Mutual-information estimator, logarithm base, confidence interval, and bias correction are declared before evaluation. The final result is a profile, not a single winner: Equation 27 cannot replace separately reported accuracy, calibration, out-of-distribution generalization, sample efficiency, latency, memory, energy proxy, equivariance, causal sensitivity, and drift.

7.5 Domains, units, normalization, and uncertainty

The architecture tuple in Equation 2 becomes numerical only after each field has a declared type. For architecture $\mathcal{A}$, let $\mathbf{X}_{\mathcal{A}}$, $\mathbf{M}_{\mathcal{A}}$, and $\mathbf{C}_{\mathcal{A}}$ denote its fast, persistent, and relational state spaces; $\mathbf{Y}_{\mathcal{A}}$ its observation space; $\mathbf{A}_{\mathcal{A}}$ its action space; and $\mathbf{O}_{\mathcal{A}}$ its output space. The update therefore has the typed signature

$$F_{\mathcal{A}} : (\mathbf{X}_{\mathcal{A}} \times \mathbf{M}_{\mathcal{A}} \times \mathbf{C}_{\mathcal{A}}) \times \mathbf{Y}_{\mathcal{A}} \times \mathbf{A}_{\mathcal{A}} \longrightarrow (\mathbf{X}_{\mathcal{A}} \times \mathbf{M}_{\mathcal{A}} \times \mathbf{C}_{\mathcal{A}}) \times \mathbf{O}_{\mathcal{A}}. \quad (28)$$

A translation Φ_k is defined only on the operation-specific subspace it maps; it is not silently extended to an entire architecture. If a source intervention has no target operation with compatible input, state, temporal support, and endpoint, the comparison is recorded as `not-comparable`, not assigned a preservation score of zero.

Physical variables retain physical units until a training-only encoding map is frozen. Position is recorded in metres or declared synthetic-world units; orientation and phase in radians; frequency in hertz; delay in seconds or samples with the sampling interval; voltage in volts; and amplitude, power, and imaging signals in their instrument units. Machine latent coordinates are dimensionless unless the generator supplies a calibrated physical map. For a noncircular scalar feature x_k , the dimensionless encoding is

$$\tilde{x}_k = \frac{x_k - \hat{a}_{k,\text{train}}}{\max(\hat{b}_{k,\text{train}}, \epsilon_b)}, \quad \epsilon_b > 0, \quad (29)$$

where $\hat{a}$ is a preregistered location estimator and $\hat{b}$ a scale estimator fit without validation or test data. A feature with zero training scale is marked constant and removed or separately coded; it is not divided by an arbitrary test-set variance.

Phase is circular. Within route and context stratum c , the tonic phase estimator is

$$\hat{\mu}_{\phi,i,c} = \text{Arg} \left(\sum_{t \in \mathcal{T}_{\text{train},c}} e^{i\phi_i(t)} \right), \quad \delta_{\phi,i}(t) = \text{wrap}_{[-\pi,\pi]}[\phi_i(t) - \hat{\mu}_{\phi,i,c}], \quad (30)$$

so the arithmetic mean does not create a false discontinuity at $-\pi/\pi$. Frequency, amplitude, duration, sign, and receiver-state components use their own estimators before they are encoded into a product-space record.

The operation-preservation score in Equation 8 is calculated only after a metric E_k , an admissible error margin $\varepsilon_k > 0$, and a replication unit are frozen:

$$P_k = \max \left(0, 1 - \frac{E_k}{\varepsilon_k} \right), \quad E_k \geq 0. \quad (31)$$

Results report E_k , ε_k , P_k , and a cluster- or seed-level confidence interval rather than the score alone. Weights w_k are fixed before the sealed test, sum to one, and are sensitivity-tested; no conclusion may depend on one convenient weighting without disclosure.

7.6 Identifiability and limiting cases

Receiver state earns a place only through incremental and intervention-sensitive value. Let $\mathcal{L}_0$ be held-out loss for the strongest frozen baseline and $\mathcal{L}_M$ the loss after adding persistent receiver state under the same

information and compute contract:

$$\Delta_M = \mathcal{L}_0 - \mathcal{L}_M. \quad (32)$$

The preregistered receiver-state claim requires a positive uncertainty-bounded Δ_M , plus a selective perturbation whose effect exceeds a matched sham:

$$\Gamma_M = \Delta J_{\text{target}} - \Delta J_{\text{sham}} > \gamma_{\min}, \quad (33)$$

with $\gamma_{\min}$ fixed before the sealed test. This separates a useful state variable from a correlated nuisance proxy.

The following limiting cases are mandatory fixtures. If recurrence is set to zero, the recurrent route must reduce to its declared feedforward update. If $M_i(t)$ is constant, the receiver-state model must reduce to a model without receiver-dependent memory. If $\tau_{ji} = 0$, delay alignment must reduce to same-step comparison. If sign is removed while magnitude is preserved, opposite excitation-like and inhibition-like effects become deliberately nonidentifiable. If all observed endpoints agree while translated lesions differ, Equation 6—not endpoint equality—determines the disposition. If a target architecture has no intervention-compatible operation, the registry must retain the residual rather than invent a metaphorical match.

A final counterexample guards against a common hidden shortcut. Suppose two episodes have identical current observations but different histories and require different outputs. Any deterministic current-observation-only model must return the same answer for both; history or receiver state is structurally required. Conversely, when no such paired episodes exist, benchmark performance cannot identify the history mechanism. The generator therefore includes observation-matched context reversals and stores their pair identifiers.

7.7 Machine-readable benchmark freeze

The human-readable design is mirrored in `application-proof/BENCHMARK-SPEC-DRAFT2.json` and `application-proof/MOVING-ASSEMBLY-WORLD-SCHEMA-DRAFT2.json`. The freeze declares eight tasks, six model families, twelve ablations, eight negative controls, disjoint seed ranges, permitted and prohibited information channels, metric identifiers, translated interventions, and the unopened final-test state. These files specify future execution; they contain no trained model, performance result, or biological claim.

7.8 Mathematical type, domain, and unit audit

The benchmark is not mathematically defined merely because every field has a name. Each field must have a domain, unit, admissible missingness rule, and comparison metric. The mathematical specification therefore freezes the following registry for all later implementations.

For orientations, the primary error is the geodesic distance

$$d_{SO(3)}(R_1, R_2) = \cos^{-1} \left[\text{clip} \left(\frac{\text{tr}(R_1^T R_2) - 1}{2}, -1, 1 \right) \right]. \quad (34)$$

The clipping is numerical protection, not a change in geometry. Position and orientation errors are reported separately because metres and radians cannot be added without an application-specific conversion. A composite pose loss is permitted only after the conversion weights and their operational meaning are frozen.

Quantity	Domain	Unit or coordinate convention	Required edge-case rule
position p	$\mathbb{R}^3$	metres or declared synthetic-world units	convert before comparison; never average mixed units
orientation R	$SO(3)$, or unit quaternions modulo $q \sim -q$	geodesic radians	normalize quaternions; clip inverse-cosine arguments
phase ϕ	circle S^1	radians in $[-\pi, \pi)$	undefined when amplitude or resultant length is below a frozen threshold
frequency f	$\mathbb{R}_{\geq 0}$	hertz	phase is not inferred at zero frequency without an explicit oscillator model
delay τ	$\mathbb{R}_{\geq 0}$ or $\mathbb{Z}_{\geq 0}$	seconds, or samples plus sampling interval Δt	sample and time delays cannot be mixed implicitly
amplitude or power	instrument-specific subset of $\mathbb{R}$	volts, squared volts, fluorescence units, or declared proxy	log ratios require positive values and a positive reference
signed effect σ	$\{-1, 0, +1\}$, or a preregistered continuous signed domain	categorical sign, or declared physical unit	zero and missing are different states
latent state z	$\mathbb{R}^d$ or a declared manifold	dimensionless unless calibrated	dimension and gauge freedom must be reported
categorical probability π	simplex Δ^{K-1}	dimensionless	entries must be finite, nonnegative, and sum to one within tolerance
typed relation ρ	finite declared relation set or graph	relation-specific	empty admissible sets are not-applicable, not perfect scores

7.9 Circular estimators, sampling, and a phase-boundary example

For phases $\phi_1, \dots, \phi_n$ within one training-only route and context stratum, define the mean resultant vector and length

$$\bar{c} = \frac{1}{n} \sum_{j=1}^n \cos \phi_j, \quad \bar{s} = \frac{1}{n} \sum_{j=1}^n \sin \phi_j, \quad \hat{R} = \sqrt{\bar{c}^2 + \bar{s}^2}, \quad \hat{\mu}_\phi = \text{atan2}(\bar{s}, \bar{c}). \quad (35)$$

The mean direction is reported only when $n \geq n_{\min}$ and $\hat{R} \geq R_{\min} > 0$, with both thresholds frozen before evaluation. When $\hat{R}$ is near zero, the observations may be broadly dispersed or multimodal; a single mean direction is then not identifiable. The record retains dispersion or a preregistered mixture estimate rather than inventing a tonic phase.

Receiver-relative phase error uses

$$d_{S^1}(\phi, \mu) = |\text{atan2}(\sin(\phi - \mu), \cos(\phi - \mu))| \in [0, \pi]. \quad (36)$$

Worked phase example. Let the expected phase be 179° and the observed phase be -179° . Arithmetic subtraction gives -358° , a false near-full-circle error. Equation 36 gives $+2^\circ = 0.0349$ radians. A system that fails this fixture cannot validly estimate a phase residual or PWD phase component.

Sampling must support the claimed temporal variable. For a declared maximum resolvable frequency $f_{\max}$, the minimum mathematical condition is

$$f_s > 2f_{\max}, \quad (37)$$

and the implementation must preregister an anti-alias filter and a stricter engineering margin. A benchmark

cannot infer a 100 Hz phase relation from a 150 Hz sample stream merely because the downstream model accepts the samples. Delay in samples is converted by $\tau = k\Delta t$; fractional-delay interpolation, windowing, and filter group delay must be fixed and included in the uncertainty analysis.

7.10 Estimator definitions and leakage-resistant worked examples

Equation 29 requires concrete estimators. The default scalar encoding uses the training median for $\hat{a}$ and the training median absolute deviation multiplied by 1.4826 for $\hat{b}$, unless a task freezes a different estimator. A constant feature with $\hat{b} = 0$ is flagged and removed or represented by a separate constant-state indicator. The ϵ_b safeguard prevents numerical division by zero; it does not manufacture variation.

Worked normalization example. If the training location and scale are 10 mV and 2 mV, a 13 mV observation becomes 1.5. Re-estimating the scale as 3 mV from the test set would change the same observation to 1.0 and leak test-distribution information. The latter value is inadmissible even if it improves the final score.

For discrete benchmark targets, the primary information readout is a held-out cross-entropy gain rather than an unqualified plug-in mutual-information number. If $\mathcal{L}_{\text{null}}$ and $\mathcal{L}_Z$ are mean natural-log losses from a frozen null decoder and a decoder given representation Z , respectively, then

$$\widehat{\Delta I}_{\text{heldout}} = \frac{\mathcal{L}_{\text{null}} - \mathcal{L}_Z}{\ln 2} \quad \text{bits per held-out trial.} \quad (38)$$

Both decoders use the same training split, target support, tuning allowance, and evaluation trials. A positive value is evidence of additional predictive information under that decoder class, not proof that Z is a unique biological code. Continuous-target information requires a separately frozen estimator and simulation-based bias audit; it cannot silently reuse Equation 38.

Equation 38 is deliberately decoder-relative and can be negative on held-out data. It is not silently substituted for the nonnegative population quantity in Equation 27. Any efficiency report must name its numerator as either an audited mutual-information estimate or the signed held-out log-loss gain, retain negative values, and show the unnormalized numerator beside the cost denominator.

For matched, column-centered representation matrices $X \in \mathbb{R}^{n \times p}$ and $Y \in \mathbb{R}^{n \times q}$, the specification defines linear CKA as

$$\text{CKA}(X, Y) = \frac{\|X^T Y\|_F^2}{\sqrt{\|X^T X\|_F^2 \|Y^T Y\|_F^2}}. \quad (39)$$

If either denominator factor is zero, CKA and Equation 26 are undefined, not zero. Episode identities must match row by row. If an alignment is learned, it is fit on training identities only and evaluated on held-out identities; a test-fitted alignment can erase genuine readout failure.

For K -class predictions, probability calibration is primarily reported by negative log loss and the Brier score

$$\mathcal{B} = \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K (\hat{\pi}_{ik} - \mathbf{1}[y_i = k])^2. \quad (40)$$

Expected calibration error may be included as a secondary sensitivity statistic because it depends on binning. Invalid probability vectors fail validation before any calibration score is computed.

Equation 22 is evaluated in a dimensionless task metric after each physical component has been reported in its native unit and scaled by a training-frozen tolerance. In Equation 23, categorical relations use zero-one or preregistered cost-sensitive loss; continuous pose relations use their native geodesic or displacement error and a frozen clipping tolerance. Equation 24 uses the task’s declared proper loss and is **not-applicable** when $T < 2$. Aggregation is first within an episode, then within an independent seed or biological unit, and only then across units; pooling frames cannot change the replication count.

Worked preservation example. For $E_k = 0.03$ and $\varepsilon_k = 0.05$, Equation 31 gives $P_k = 0.40$. For $E_k = 0.07$, it gives zero. Neither value is computed when the source and target operations lack compatible domains or interventions; that case is **not-comparable**, not a low score.

7.11 Identifiability, causal contrasts, and explicit counterexamples

Incremental prediction and selective perturbation answer different questions. Equation 32 tests whether receiver state improves held-out prediction under a matched contract. Equation 33 tests whether intervening on that state has a selective consequence. Both use paired seeds or paired experimental units. For seed or unit r , define

$$\hat{\Gamma}_{M,r} = [J_{r,\text{base}} - J_{r,\text{target}}] - [J_{r,\text{base}} - J_{r,\text{sham}}] = J_{r,\text{sham}} - J_{r,\text{target}}. \quad (41)$$

Inference is performed on the paired $\hat{\Gamma}_{M,r}$ values, not on unpaired trial counts. The target and sham must match intervention duration, magnitude, missingness, and nonspecific compute or physiological disturbance. A positive Δ_M without a selective Γ_M can indicate a useful correlate, proxy, or regularizer; a selective Γ_M without held-out information can indicate necessity for execution without decodable content. Neither pattern is silently promoted to the combined claim.

A matched rescue must restore the targeted state or route while leaving the principal nonspecific disturbance in place. Recovery after removing the entire perturbation is not a rescue. In biological data, off-target effects, common input, motion, arousal, and intervention-induced rate or power changes remain explicit covariates and negative controls; predictive adjustment alone does not prove that all causal confounding has been removed.

Context-reversal counterexample. Consider balanced paired episodes (A, B) with identical current observation y_t but opposite required binary outputs because their histories differ. A deterministic model $h(y_t)$ must make the same prediction on both. Its error over each balanced pair is therefore bounded by

$$\text{err}[h(y_t)] \geq \frac{1}{2}. \quad (42)$$

A history-aware model can escape this bound, but only if the history channel is not inferable from an accidental current-frame artifact, episode identifier, padding pattern, or random seed. The generator must test equality of the declared current observations byte-for-byte and run a nuisance classifier against forbidden metadata.

Exact translated-operation example. Let architecture A have scalar state s , action a , and update $F_A(s, a) = s + a$. Let architecture B have $u = 2s$, translated action $b = 2a$, and update $F_B(u, b) = u + b$. With $\Phi(s) = 2s$ and $\phi(a) = 2a$,

$$\Phi(F_A(s, a)) = 2(s + a) = 2s + 2a = F_B(\Phi(s), \phi(a)). \quad (43)$$

If both systems use the state-clamp intervention $I(x) = 0$, that intervention also commutes with Φ . This is an exact operation-level equivalence over the declared scalar subspace; it does not imply equivalence of the complete architectures.

Endpoint-only counterexample. Suppose every observed evaluation episode has positive final state. The dynamic system in Equation 43 and a constant classifier that always returns “positive” can have identical endpoint accuracy. Under an action reversal or state clamp, the dynamic system changes as declared and the constant classifier does not. Endpoint agreement therefore cannot replace Equations 5 and 6.

PWD identifiability uses a locked comparator ladder: rate; rate plus power; phase; raw finite history; recurrent state; reference frame; persistent receiver state; and receiver-relative residual. Every added model receives the same permitted observation history and budget. The PWD residual is not identified if it can be reconstructed from a prohibited episode key, from a normalizer fit on evaluation data, or from a target-derived phase reference. Removing sign while preserving magnitude is a deliberate negative control: if excitation-like and inhibition-like routes require different predictions, a magnitude-only model must lose that distinction.

7.12 Drift, gauge freedom, and readout stability

Representations can move without losing all function. Let a row-vector code transform by an orthogonal matrix Q : $Z' = ZQ$, with $Q^T Q = I$. Then $Z' Z'^T = Z Z^T$, so the linear CKA in Equation 39 is one when the nondegeneracy conditions hold. Yet a fixed linear decoder w can fail after the rotation. The transported decoder

$$w' = Q^T w \quad \text{satisfies} \quad Z' w' = Z Q Q^T w = Z w. \quad (44)$$

Worked drift example. Let $z = (1, 0)$, $w = (1, 0)^T$, and let Q rotate codes by 90 degrees. The original readout is one. The new code is $(0, 1)$, so the unmodified decoder returns zero, while the transported decoder returns one. Pairwise geometry can therefore remain stable while a fixed biological or machine readout loses access. the benchmark consequently requires four separate reports: raw-code similarity, fixed-decoder performance, training-aligned decoder performance on held-out identities, and behavior.

This example also supplies a counterexample to an overstrong drift claim. High CKA does not prove functional access by a fixed receiver. Conversely, low coordinate correlation need not prove loss of relational content when an admissible alignment or downstream adaptation preserves behavior. Alignment complexity and its training data are part of the mutable-state and compute budgets; an arbitrarily expressive test-fitted map is prohibited.

7.13 Stability and timescale checks

The SAN condition contains fast receiver state q_t and relatively persistent structure M_t . “Two timescales” is an empirical model condition, not a verbal label. Let τ_q be the preregistered relaxation or autocorrelation time of the fast state and τ_M that of the persistent state. Define

$$\kappa = \frac{\tau_M}{\tau_q}. \quad (45)$$

The paper may use a separated-timescale approximation only when the lower confidence bound for κ

exceeds a preregistered threshold $\kappa_{\min} > 1$ and sensitivity analysis shows that conclusions persist across an admissible range. If the timescales overlap, the implementation uses a coupled-state model and reports that the separation was not supported. No universal biological value of κ is assumed.

For an autonomous differentiable discrete update $q_{t+1} = F(q_t; M)$ with fixed point q^* , local asymptotic stability is supported when

$$\rho\left(\left.\frac{\partial F}{\partial q}\right|_{q^*}\right) < 1, \quad (46)$$

where ρ is spectral radius. This is a local result, not a guarantee for driven, switched, saturated, or stochastic dynamics. For a driven contraction with state Lipschitz constant $L_q < 1$ and input gain L_u , perturbations obey the bound

$$\|\delta q_t\| \leq L_q^t \|\delta q_0\| + L_u \sum_{j=0}^{t-1} L_q^{t-1-j} \|\delta u_j\|. \quad (47)$$

When $L_q \geq 1$, a finite-horizon empirical stability report may still be possible, but contraction cannot be claimed. Iterative capsules, recurrent columns, transformer recurrence, and the SAN condition all receive the same checks over their declared update horizon: bounded state norm, finite outputs, perturbation amplification, convergence or stopping condition, and sensitivity to initialization.

The minimum timescale ledger contains: sample interval Δt ; feature-extraction window; route delay τ_{ji} ; recurrent iteration interval; episode duration; persistent-state update interval; training/plasticity interval; and drift interval. A result is invalid if a “persistent” feature is estimated from a future test window, if an event window straddles the target outcome, or if filter delay is mistaken for neural or computational delay.

Mandatory edge fixtures include: $T = 1$, for which Equation 24 is not applicable; zero recurrence; constant M ; zero and maximum declared delay; empty relation set; no-action episodes; vanished signal amplitude with undefined phase; rotations separated by π ; antipodal quaternion representations of the same rotation; saturated activation; all-missing channel; variable object count; and a perturbation at the first and last legal time step. Passing the ordinary task does not substitute for these fixtures.

7.14 Statistical power, replication units, and multiplicity

The replication unit is the independently initialized training seed for machine comparisons and the animal or other independently sampled biological unit for biological inference, unless a hierarchical design justifies another level. Frames, trials, neurons, or patches nested inside one seed, animal, or session do not become independent replicates by being numerous. All confidence intervals and resampling procedures preserve the declared nesting.

For an approximate paired normal comparison with standardized mean effect d , two-sided Type-I error α , and power $1 - \beta$, the initial planning bound is

$$n_{\text{pair}} \geq \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{d} \right)^2. \quad (48)$$

Worked planning example. With $d = 0.5$, $\alpha = 0.05$, and 80% power, the normal approximation gives

$(1.96 + 0.84)^2 / 0.5^2 = 31.36$, rounded up to 32 independent pairs. A 20% attrition reserve raises recruitment to 40. This is a planning illustration, not the final sample size for this paper: nonlinear metrics, clustered biological observations, missingness, seed variance, and multiple primary contrasts require simulation from pilot or conservative variance assumptions.

Claims of practical equivalence use a preregistered margin $\delta > 0$, not a nonsignificant difference. For effect θ , equivalence is supported only when the chosen $(1 - 2\alpha)$ confidence interval lies wholly inside the margins:

$$[L_\theta, U_\theta] = \text{CI}_{1-2\alpha}(\theta), \quad -\delta < L_\theta \leq U_\theta < \delta. \quad (49)$$

The benchmark freezes one primary endpoint per hypothesis family or applies a declared multiplicity procedure. Exploratory metrics remain labeled exploratory. Paired bootstrap or permutation tests resample seeds or independent units, not individual frames. For the biological bridge, power simulation must reproduce the nesting of animals, sessions, neurons, and trials, and must include the planned perturbation and matched-sham contrast.

7.15 Compute, memory, interaction, and energy-proxy budgets

Fairness is a vector constraint, not equal parameter count alone. For model m and task t , let A_{mt} , C_{mt} , and R_{mt} be the numbers of applicable ablations, controls, and rescue conditions. With S_{mtc} independent seeds in condition c , the complete run count is

$$N_{\text{runs}} = \sum_{m=1}^6 \sum_{t=1}^8 \sum_{c=1}^{1+A_{mt}+C_{mt}+R_{mt}} S_{mtc}. \quad (50)$$

The base grid alone contains $6 \times 8 \times S$ runs when all cells use S seeds. With five seeds, that is 240 base runs. An illustrative four additional eligible conditions per cell would produce 1,200 runs. The actual count must be generated from the frozen eligibility matrix; nonapplicable perturbations are not fabricated merely to equalize row counts.

Total training, tuning, and evaluation compute are reported separately:

$$C_{\text{total}} = \sum_r (C_r^{\text{train}} + C_r^{\text{eval}} + C_r^{\text{tune}}), \quad M_{\text{peak}} = \max_r M_r, \quad L_{p95} = \text{quantile}_{0.95}(L_r). \quad (51)$$

The ledger also records parameter count, mutable-state bytes, training examples, observation tokens or sensor frames, action opportunities, recurrent iterations, wall time, hardware class, peak memory, and measured or proxy energy. FLOPs, wall time, GPU-hours, and joules are different quantities. A hardware-independent operation estimate cannot be relabeled as measured energy; where power telemetry is unavailable, the paper reports an energy proxy and says so.

No single weighted efficiency score determines the winner. Model a Pareto-dominates model b only if it is no worse on every preregistered benefit and cost coordinate and strictly better on at least one after uncertainty is considered:

$$a \succ_P b \iff \bigwedge_j B_j(a) \geq B_j(b) \wedge \bigwedge_k C_k(a) \leq C_k(b) \wedge \text{at least one strict inequality}. \quad (52)$$

Scaling curves are reported across at least three feasible budget points before making an efficiency claim. A model that exceeds the agreed memory, data, interaction, or tuning budget is a budget violation, not a failed scientific mechanism; it is reported separately and can be studied in an explicitly unequal-budget analysis.

7.16 Mathematical decision rules

The following rules govern all future empirical outputs:

1. **Typed before compared.** A quantity with no declared domain, unit, missingness rule, and metric is not comparable.
2. **Estimable before scored.** Circular degeneracy, zero variance, empty relations, invalid probabilities, and test-fitted preprocessing yield undefined or **not-applicable**, not convenient numeric values.
3. **Predictive and causal evidence remain separate.** Incremental held-out value and selective perturbation are both required for the combined receiver-state claim.
4. **Endpoint equality is insufficient.** An operation match requires the execution and intervention diagrams to commute within frozen margins.
5. **Drift is readout-relative.** Geometry, fixed-decoder access, aligned access, and behavior are reported separately.
6. **Stability is local to a stated regime.** A spectral-radius or contraction result cannot be generalized beyond its assumptions and horizon.
7. **The independent unit controls uncertainty.** Nested trials do not inflate seed-, session-, or animal-level sample size.
8. **Equivalence requires an equivalence test.** Failure to reject a difference is not evidence that two mechanisms are the same.
9. **Budgets remain plural.** Parameters, compute, state, data, interaction, latency, memory, and energy proxy are all retained.
10. **Counterexamples are fixtures.** Every implementation must pass the phase-wrap, current-observation ambiguity, endpoint-only match, drift/readout, zero-variance, and translated-intervention fixtures before the sealed benchmark may open.

7.17 Executable development gate

The mathematical contract now has an executable development layer at the accompanying executable development package. The generator admits only train, validation, and development-test seed ranges. It rejects all final-test seeds before scene construction. Its output records retain the frozen episode, scene-graph, observation, action, counterfactual, intervention, and provenance structure while the semantic validator additionally checks finite values, exact field sets, vector arity, unit quaternions, frame acyclicity, graph references, split/seed consistency, allowed action forms, forbidden input fields, translated targets, and content hashes.

Nine deterministic development records cover all eight tasks: one record for T1, T2, T3, T4, T5, T7, and T8, plus both histories of a T6 context-reversal pair. The T6 validator checks that scene graph, action sequence,

and current observation match exactly while earlier observations differ and pair identifiers point to each other. These are synthetic contract fixtures, not a dataset of sufficient size for model evaluation.

The executable mathematical fixtures include the circular phase-boundary example, degenerate circular mean, constant-feature normalization rejection, held-out information gain, probability-simplex and Brier checks, preservation margins, exact translated-operation commutation, and an endpoint-only countermodel. The countermodel agrees with the dynamic model on an ordinary positive endpoint but diverges under action reversal and state clamp. This demonstrates in code why endpoint equality does not establish operation preservation; it does not establish that any two research architectures are equivalent or non-equivalent.

The stability layer distinguishes a linear fixture with spectral radius 0.5 and infinity-norm gain 0.6 from an unstable fixture with spectral radius approximately 1.1. Sampling fixtures distinguish 1,000 Hz sampling of a 100 Hz maximum from the inadmissible 150 Hz case. Timescale fixtures distinguish a point-estimate ratio of 100 from a ratio of 1.2 under a development threshold of 5. The timescale report explicitly retains the result-level requirement for an uncertainty-bounded ratio; the fixture cannot substitute for empirical estimation.

Twenty-five tests now pass: twelve generator, estimator, monitor, and budget unit tests; nine adversarial semantic-validator tests; and four integration, provenance, manifest, and seal tests. The test ledger also retains the initial run in which 22 tests passed and three integration tests caught four lowercase JSON booleans mistakenly used in Python dictionaries. The correction was confined to the seal writer, and the final test was never materialized or opened.

The final-test plan declares seeds 7000–7999 and a selection rule that would eventually produce 9,000 episode records: eight tasks plus a second T6 history for each of 1,000 seeds. The executable development stage materializes no final-test episode and no final-test episode hash. The seal state is `sealed_unopened`, unseal count is zero, no opened timestamp exists, and result authorization is false. Only the plan and seal files are permitted in the final-test directory.

The locally installed third-party JSON Schema package remains permission-blocked, so The executable development stage does not claim external JSON Schema 2020-12 meta-schema execution. It checks the frozen schema’s dialect and structure and runs an independent strict typed/semantic validator on every generated fixture. A hermetic external-schema environment remains an implementation improvement, not a hidden completed check.

This gate establishes that the development generator and validators execute their declared fixtures. It reports no trained M0–M5 model, task score, PWD advantage, architecture ranking, biological effect, model equivalence, or final-test result.

7.18 Capacity-matched synthetic development/validation pilot

The next implementation problem is simple to state. A benchmark can appear to distinguish mechanisms merely because one model received more features, more fitted coefficients, more training examples, a different optimizer, or access to a privileged variable. Before spending the substantially larger budget required by faithful end-to-end architectures, the measurement pipeline therefore needs a controlled test in which capacity and data access are equal but the permitted transformations differ in declared ways.

The capacity-matched pilot implements six compact **operational substitutes**, not full cortical, Hawkins-style, capsule, transformer, or SAN systems. Every substitute maps a synthetic episode to exactly eight real-valued features,

$$\phi_m^{(c)} : \mathcal{X}_{T,s} \rightarrow \mathbb{R}^8, \quad (53)$$

where $m \in \{M0, \dots, M5\}$, (c) is a frozen baseline, ablation, rescue, or negative-control condition, (T) is the task, and (s) is an allowed train, validation, or development seed. A deterministic logistic readout uses one bias plus eight weights,

$$\hat{p}(y = 1 \mid x, m, c) = \sigma\left(w_0 + w^\top \phi_m^{(c)}(x)\right), \quad (54)$$

and minimizes the same finite-sample objective for 60 full-batch epochs,

$$\hat{L}(w) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)] + \lambda \sum_{j=1}^8 w_j^2, \quad n = 64, \lambda = 0.01. \quad (55)$$

All conditions use the same 80-case training candidate pool per task, the same deterministic 64-case subset rule, 20 validation cases, 40 development cases, learning rate 0.15, epochs, optimizer, threshold, and five initialization seeds (11, 29, 47, 71, and 101). The label generator includes a frozen 0.08 deterministic flip probability. No hyperparameter is selected from development performance. M0–M5 each have eight features and nine trainable coefficients. Their declared mutable-state proxies remain different—0, 2, 2, 3, 3, and 4 scalar slots respectively—and are reported instead of being hidden inside a claim of perfect fairness. Each run has the same training operation-count proxy of 161,280 and validation-plus-development inference proxy of 1,260; these are algorithmic counts, not joule measurements.

The feature transformations isolate the intended discriminators. M0 receives current-input primitives without a mechanism-specific interaction. M1 adds recurrent/history terms; M2 adds object-relative location products; M3 adds agreement and routing products; M4 adds attention-like relation, action, drift, and route interactions; and M5 adds persistent receiver state, a fast current-minus-tonic residual, receiver-residual interaction, action return, drift alignment, and route-intervention interaction. Because these are deliberately engineered eight-feature maps with a common linear readout, their scores answer whether the frozen pipeline detects supplied operations. They cannot tell us how a production transformer, biological cortical circuit, or complete SAN architecture would perform.

The run grid contains 240 baseline rows (six substitutes by eight tasks by five seeds) and 135 condition rows (27 frozen intervention cells by five seeds), for 375 retained rows. No run failed. Mean development accuracy across all eight tasks was 0.493 for M0, 0.598 for M1, 0.700 for M2, 0.631 for M3, 0.860 for M4, and 0.753 for M5. These macro averages are descriptive only because each substitute was intentionally given a different subset of task-relevant transformations; they are not a league table and cannot rank the named research architectures.

The six preregistered development contrasts behaved as the task construction predicts. Relative to M0, M3 improved T1 relation recovery by 0.495 (five-seed 95% t interval 0.455 to 0.535); M2 improved T2 pose-location transfer by 0.160 (0.104 to 0.216); M1 improved T3 history-dependent occlusion by 0.410 (0.368 to 0.452); M4 improved T5 action-conditioned selection by 0.520 (0.480 to 0.560); and M5 improved T6 observation-matched context reversal by 0.360 (0.258 to 0.462) and T8 route-intervention prediction by 0.355 (0.288 to 0.422). These intervals quantify variation across five initialization seeds under one deterministic data generator. They are neither population confidence intervals nor evidence for biology.

The causal checks are more informative than the raw scores. Removing recurrence reduced M1 by 0.405 on T3 and 0.380 on T6. Removing the reference-frame interactions reduced M2 by 0.215 on T2 and 0.370 on T7.

Removing agreement/routing terms reduced M3 by 0.450 on T1, 0.525 on T4, and 0.370 on T8. Equalizing the task-relevant attention-like interactions reduced M4 by 0.255 on T1, 0.510 on T5, and 0.370 on T7. For M5, erasing receiver state reduced T3 by 0.460 and T7 by 0.355; removing action feedback reduced T5 by 0.585; removing recurrence reduced T6 by 0.340; lesioning the route-intervention operator reduced T8 by 0.345; and removing negative-signed values reduced T8 by 0.045. A matched T8 rescue restored the baseline exactly because this control reinstates the baseline feature map, whereas a mismatched rescue remained at the lesion level, 0.345 below baseline. The matched result is therefore a deterministic translation-and-plumbing check, not evidence that a model learned self-repair. Label shuffles, history shuffles, and nuisance-only controls all reduced their target-cell scores. Removing an unrelated component produced a zero mean change in its two frozen control cells, as intended.

Negative-result retention changes the interpretation. Twenty-four of the 48 baseline model-task cells had mean development accuracy at or below 0.55. M0 was near chance or worse on seven tasks; M1 on five; M2 on four; M3 on four; M4 on one; and M5 on three. These rows are not discarded as irrelevant failures. They demonstrate that the compact maps are selective and incomplete, and that supplying one useful interaction does not create a generally capable architecture. No ablation unexpectedly improved its cell, no negative control produced a gain above 0.05, and no execution failure occurred; the complete row-level ledger remains available rather than only the favorable contrasts.

For each primary contrast, the pilot also records the paired effect

$$\hat{\Delta}_{m,T} = \frac{1}{5} \sum_{s \in S} \left(\hat{A}_{m,T,s} - \hat{A}_{M0,T,s} \right), \quad (56)$$

its seed-level standard deviation, and an exploratory normal-approximation sample-size calculation. The large engineered effects yield a nominal five-pair planning floor for all six contrasts. That number is not accepted as the final-test sample size: initialization seeds are not animals, people, independent laboratories, or natural task populations; the generator deliberately exposes the contrast; and no final multiplicity procedure has been run. The power artifact is therefore a freeze and planning receipt, not authorization to inspect the final split.

The executable pilot verifies the final-test plan and seal before and after all 375 runs. Their hashes remain byte-identical, the seal remains `sealed_unopened`, unseal count remains zero, no episode content or hash has been materialized, and result authorization remains false. The environment lock records a standard-library-only Python implementation, no network access, no accelerator dependency, and exact source hashes. Eighteen Draft 5 tests cover contract capacity, split isolation, observation-matched T6 pairs, feature arity, deterministic sampling and fitting, interventions, rescue logic, signed-value handling, resource accounting, full-grid retention, negative results, power scope, manifest integrity, and final-seal preservation.

The bounded conclusion is therefore procedural. The synthetic pilot shows that the typed application can distinguish deliberately supplied operations under equal feature, coefficient, data, epoch, and optimizer budgets; it also exposes where every compact substitute fails. It does not validate PWD in cortex, demonstrate that M5 is superior to a production transformer, establish equivalence between any architectures, or answer the sealed benchmark. Draft 6 must decide whether to replace these substitutes with faithful capacity-matched implementations or retain them as pipeline controls while adding full models. Only a separately authorized, preregistered final run may convert the sealed split into confirmatory evidence.

7.19 Mechanism-bearing reference calibration

The capacity-matched pilot answered a narrow question: can the measurement pipeline detect a supplied interaction when every compact control has the same feature and readout capacity? It could. That success creates a harder problem. A hand-selected product such as pose times location can make the right task easy without reproducing the mechanism by which a reference-frame model, capsule system, transformer, cortical circuit, or SAN receiver would obtain and update that relation. Calling those eight-feature controls the original architectures would compress each theory into a strawman.

The full originals cannot be instantiated honestly in one uniform local program. Biological cortical circuits have no single canonical executable implementation and cannot be reduced to a software checkpoint without experimental data. Hawkins-style, capsule, and transformer families each contain multiple incompatible implementations and training regimes. The SAN receiver-state model is itself an experimental proposal, not a validated biological simulator. The mechanism-bearing calibration therefore uses a **two-tier design**. The complete C0–C5 compact-control record from The capacity-matched pilot remains immutable. A new R0–R5 tier implements more of each proposed operation while retaining the label **mechanism-bearing reference prototype**. It does not use “faithful original,” “full architecture,” or “production model.” A negative R-row can falsify the adequacy of that prototype; it cannot be used against the richer named research program.

For an episode (x), initialization seed (s), reference model (m), and frozen condition (c), the reference tier computes

$$z_{m,s}^{(c)}(x) = E_{m,s}^{(c)}(x) \in \mathbb{R}^{16}, \quad \hat{p}(y = 1 \mid x) = \sigma\left(\beta_0 + \beta^\top z_{m,s}^{(c)}(x)\right). \quad (57)$$

Only the 17-coefficient readout is fitted. Encoder configuration is fixed by model identity and initialization seed, hashed, and retained with the training-case-ID hash and fitted coefficients in a 485-record checkpoint ledger. Freezing the encoder makes the mechanism and intervention trace auditable; it also limits performance and is not equivalent to end-to-end learning. Three baselines expose that limitation: B0 is the training-majority constant; B1 is a 17-parameter raw-input logistic readout; and B2 augments the 16 raw inputs with all 120 pair products, yielding 137 fitted coefficients. B2 is a higher-capacity nonlinear calibration baseline, not a capacity-matched comparator.

The reference operations are explicit:

1. **R0 feedforward reference:** one fixed (16×16) orthogonalized dense projection and tanh nonlinearity.
2. **R1 recurrent reference:** four semantically declared tokens enter an eight-state recurrence whose transition matrix has spectral radius 0.75; the readout receives final and time-averaged states.
3. **R2 sensation-location consensus:** part, pose-location, drift-location, action, and route votes undergo two declared consensus rounds.
4. **R3 capsule reference:** four primary four-vectors vote through fixed transforms to two output capsules using up to three routing-by-agreement iterations; votes, coupling matrices, and output capsules are logged.
5. **R4 attention reference:** four four-dimensional tokens enter two scaled dot-product attention heads, a frozen output projection, and a residual connection; both (4×4) attention matrices are logged.
6. **R5 receiver-state reference:** persistent state, tonic expectation, signed current-minus-tonic residual, action return, a route-intervention gate, and three recurrent consequences are separately logged.

The learned-readout objective remains common,

$$\hat{L}(\beta) = -\frac{1}{64} \sum_{i=1}^{64} [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)] + 0.01 \sum_j \beta_j^2, \quad (58)$$

with 100 deterministic full-batch epochs, learning rate 0.12, the same five initialization seeds, 20 validation cases, and 40 development cases. The development cases are separated into four disjoint ten-case blocks; observation-matched T6 history pairs never cross a block boundary. This creates two technical variation axes—encoder/readout initialization and held-out seed block—without pretending that either is a biological population replicate.

7.19.1 Baselines and mixed primary contrasts

Across eight tasks, descriptive macro development accuracy was 0.503 for B0, 0.609 for B1, 0.705 for the larger B2, 0.612 for R0, 0.592 for R1, 0.884 for R2, 0.696 for R3, 0.600 for R4, and 0.635 for R5. These averages are not a leaderboard. R2 explicitly receives the synthetic relations used by several task rules, while R1, R3, R4, and R5 expose different fixed transformations. B2 fits 137 coefficients, compared with 17 for B1 and every reference readout. Parameter, fixed-encoder, state, operation, and memory budgets remain separate in the resource ledger.

The six directional contrasts were frozen before reference result readback. Their paired technical-cell means and descriptive 95% intervals were:

Frozen development contrast	Mean accuracy difference	Descriptive 95% interval
R3 minus B1 on T1	+0.300	+0.224 to +0.376
R2 minus B1 on T2	+0.205	+0.133 to +0.277
R1 minus B1 on T3	-0.050	-0.092 to -0.008
R4 minus B1 on T5	-0.025	-0.077 to +0.027
R5 minus B1 on T6	+0.005	-0.023 to +0.033
R5 minus B1 on T8	+0.205	+0.147 to +0.263

The R1/T3 and R4/T5 directional contrasts are unfavorable and remain frozen in the negative ledger. R5/T6 is effectively unresolved at this scale. Draft 6 therefore cannot claim that its fixed recurrent encoder beats a raw linear readout on T3, that its fixed attention encoder beats B1 on T5, or that receiver state adds a reliable B1-relative advantage on T6. Positive R2, R3, and R5/T8 results show that the reference program can expose particular engineered operations; they do not establish superiority of the named theories.

7.19.2 Ablation and countercontrol readback

Several interventions behaved as intended. Resetting R1 state reduced T3 by 0.200 and T6 by 0.105. Removing R2 location channels reduced T2 by 0.260 and T7 by 0.350. Erasing R5 persistent state reduced T3 by 0.315 and T6 by 0.430. Lesioning the R5 route gate reduced T8 by 0.135, while history-shuffle and nuisance-only controls reduced T6 by 0.410 and 0.435. These are prototype-specific necessity checks.

Other cells failed the proposed necessity test. Uniform R3 routing slightly improved T4 (+0.015). Uniform R4 attention improved T7 (+0.035), and removing the R4 residual slightly improved T5 (+0.005). Zeroing R5 tonic state improved T6 (+0.010), and replacing its signed residual with an unsigned residual improved T8 (+0.035). The label-shuffled R4/T1 control improved by 0.060. Each is retained. The correct inference

is not that routing, attention, tonic expectation, sign, or labels are generally irrelevant. It is that the present fixed encoder and synthetic cell do not identify the proposed necessity, and the label-shuffle gain warns that technical variation can overwhelm a weak reference signal.

The matched R5 route rescue restores the baseline route operation by construction and returns the baseline score. The mismatched rescue remains at the lesion level. This is a translation and plumbing test, not learned recovery. A recovery claim requires a separately trained or dynamically reinstated operation whose success was not encoded by simply selecting the baseline branch.

7.19.3 Variance, multiplicity, and independence boundary

For each model-task baseline, accuracy in initialization level ($i \in \{1, \dots, 5\}$) and disjoint development block ($b \in \{1, \dots, 4\}$) is decomposed as

$$A_{ib} = \mu + u_i + v_b + \epsilon_{ib}, \quad (59)$$

using method-of-moments nonnegative components for initialization, block, and seed-by-block interaction. Averaged over tasks, block variation exceeded initialization variation for every model. For example, R5 averaged 0.00131 initialization variance, 0.01078 block variance, and 0.00739 interaction variance; R2 averaged 0.00025, 0.00451, and 0.00198. This warns against treating repeated initializations on one development set as broad replication. The decomposition describes technical sensitivity under one generator only.

The future confirmatory family remains six directional tests. If ($p_{(1)} \leq \dots \leq p_{(6)}$) are their ordered final-test values, Holm rejection requires

$$p_{(k)} \leq \frac{0.05}{7 - k} \quad \text{sequentially until the first failure.} \quad (60)$$

Secondary ablation, rescue, and translated-trace tests are assigned a Benjamini–Hochberg descriptive family at ($q=0.10$), evaluated only after the primary family. The development analysis computes no confirmatory p value. Failure to reject a difference cannot be called equivalence; equivalence requires separately frozen margins and two one-sided tests.

7.19.4 Checkpoints, translated traces, and resource vectors

Every fitted readout records

$$\mathcal{C} = (m, T, s, c, H_{\text{config}}, H_{\text{trainIDs}}, \beta, \bar{z}, \sigma_z, H_{\text{checkpoint}}), \quad (61)$$

so a result row can be joined to the exact configuration, selected training cases, standardization values, and coefficients. The 485 checkpoint records cover 360 baselines and 125 intervention fits. Seventy-three exemplar trace records cover all 48 R-model/task baselines and all 25 conditions. A translated trace registry names the internal unit, state or relation, update or route, observable, and residual that has no justified translation. Trace similarity alone never establishes an operation match.

Resource accounting reports fitted coefficients, fixed encoder scalars, state/trace scalars, iteration count, operation-count proxy, and float64 memory proxy. B1 and reference readouts each fit 17 coefficients, but

their fixed encoders and state costs differ. R3, for example, retains vote and coupling state; R4 retains token/head attention state; R5 retains persistent, tonic, residual, gate, and recurrent state. These counts are reproducibility aids. They are not physical energy measurements and cannot be converted to joules without hardware measurement.

7.19.5 Negative floor and gate decision

The mechanism-bearing calibration retains 38 new negative rows: 30 near-chance or worse model-task cells, five ablations with no observed loss, one unexpected negative-control gain, and two nonpositive primary contrasts. It also makes a byte-identical copy of all 24 pilot negative rows; both files have SHA-256 FD48F497D001C6F781A025164A4AD537D4BF6FCB1E76B81FBEA3F12CB0B91A2F. None is removed from the narrative because later code is more elaborate.

The bounded mechanism-bearing conclusion is that the application now has auditable reference mechanisms, checkpoint provenance, internal traces, independent technical variance axes, stronger baselines, and a frozen multiplicity plan. Its failures are scientifically useful: they identify which current prototypes do not yet support the proposed discriminator. The next model gate requires end-to-end learning or source-author code for the named artificial families and empirical data for the biological family. The final-test plan and seal remain byte-identical, `sealed_unopened`, with zero unseals and no episode content or hash materialized.

8 Biological bridge experiment

The computational benchmark must be paired with a biological test rather than using cortex as decorative inspiration.

Use a task in which an animal identifies an object or surface across controlled movement, partial view, and sensory conflict. Record laminar activity with high-density probes and, where feasible, cell-type- or compartment-resolved imaging. Decode sensation, object identity, pose, action, and receiver-relative residuals from held-out data. Then perform targeted perturbations that preserve simple nuisance variables as closely as possible.

The discriminator ladder is:

1. rate-only model;
2. rate plus local field power;
3. phase and phase-of-firing model;
4. laminar recurrent-state model;
5. sensation plus reference-frame model;
6. persistent receiver structure plus fast residual model; and
7. the combined model with action and returned sensory evidence.

Perturbations should include reference-frame disruption, feedback interruption, matched changes in superficial or deep laminar drive, and manipulation of receiver state while controlling mean rate and broadband power. A proposed SAN contribution is supported only when its declared residual improves held-out prediction and its selective perturbation produces the predicted failure with matched rescue. Failure of

one residual variable reopens that variable and its measurement model; it does not erase the source genealogy or force a verdict about the entire SAN architecture.

9 Hypotheses and discriminating outcomes

H1 — explicit relation advantage. Architectures with explicit pose, location, or equivariant structure will outperform equally budgeted generic baselines on viewpoint and part-whole recombination at lower data exposure.

H2 — routing is mechanism-specific. Capsule agreement, transformer attention, Hawkins lateral consensus, and SAN recurrence will show different causal-ablation signatures even when their outputs are similar.

H3 — embodiment matters selectively. Action-conditioned models will improve object-relative inference and multisensory registration more than static-image tasks; the gain should disappear when action and sensory return are decorrelated.

H4 — recurrence matters when evidence is delayed. Recurrent and iterative architectures will gain most on occlusion, temporal sequence, and delayed disambiguation, not necessarily on single-view classification.

H5 — persistent receiver state adds information. A model with learned receiver structure will outperform a state-matched model that receives only current activation when identical inputs have context-dependent consequences.

H6 — PWD adds a bounded residual. Receiver-relative timing/state departures will improve prediction beyond rate, power, raw history, and reference-frame baselines in at least one preregistered task, and selective perturbation will change the predicted route or behavior.

H7 — drift can preserve function. Internal representations may change while decoded relations and behavior remain stable; architectures will differ in how they re-register altered codes.

H8 — no universal column token is required. If column boundaries or unit definitions vary across areas, a typed interface can still compare local transformations without assuming one universal anatomical symbol.

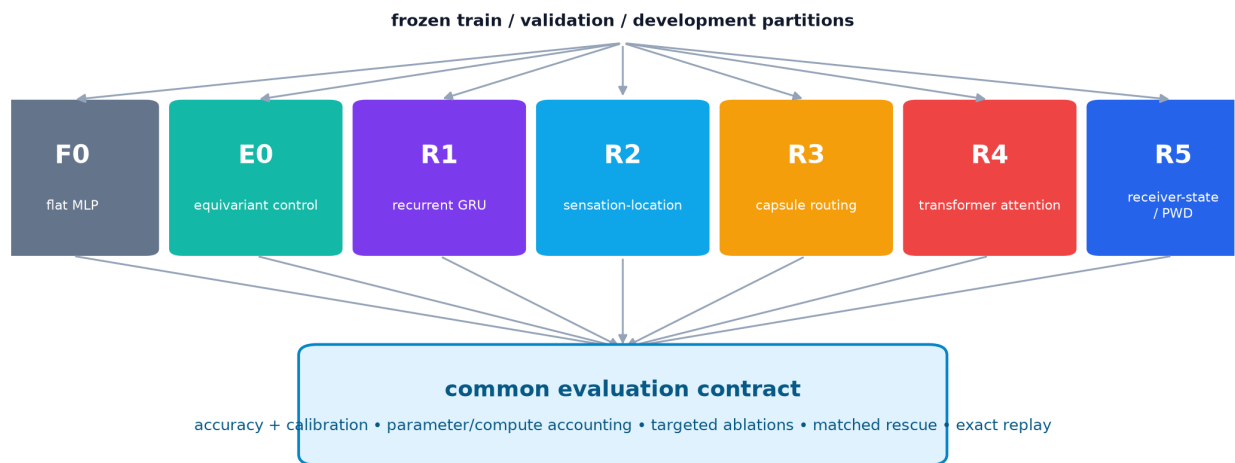
10 Trainable architecture benchmark

The fixed-encoder calibration established that the comparison could be made explicit with fixed encoders, complete traces, frozen contrasts, and preserved negative results. The remaining problem was practical. A fixed encoder can show that an operation is expressible without showing that a bounded model can learn the operation from the same data under a common capacity and training contract. The trainable application therefore replaces the fixed-encoder comparison with seven small, genuinely trainable models while preserving every earlier claim, equation, source boundary, and adverse row.

The benchmark asks a deliberately narrower question than “Which theory is best?” Given the same sixteen input scalars, eight task labels, training partition, optimizer, epoch count, and development partition, can small implementations learn the particular operations assigned to recurrence, sensation–location binding, capsule routing, attention, and SAN receiver-state/PWD processing? A flat nonlinear model and a declared token-permutation-equivariant model serve as controls. The result is an implementation audit, not a ranking of complete theories.

Figure 1. Frozen trainable architecture benchmark

The same 16-scalar scene contract and eight task labels feed seven small, genuinely trainable models.



Development-only. Small operational implementations. No biological inference. Future final seal unopened.

Figure 1: Frozen trainable benchmark architecture. Seven small model families receive the same declared sixteen-scalar interface and task label while retaining operation-specific internal structure. The diagram represents implemented benchmark roles, not complete source-author or biological architectures.

10.1 Frozen application contract

The complete contract was frozen before the scientific run. Each task supplied 80 training, 20 validation, and 40 development cases. Five initialization seeds were fixed: 11, 29, 47, 71, and 101. All models used AdamW for 80 epochs without early stopping or development-set tuning. The final seeds 7000–7999 remained forbidden, and the previously hashed final-test plan and seal remained unopened.

The parameter band was 700–8,000 trainable parameters with a maximum sixfold ratio across families. The observed counts were 729–3,873, a 5.31 ratio. The point was not exact equality, which would conceal structurally necessary parameters, but a declared small-model band paired with full parameter, checkpoint, memory, training-time, inference-time, and profiler accounting.

The seven trained systems were:

- **F0, flat control:** a two-layer nonlinear model over the declared flat input and task embedding;
- **E0, equivariant control:** a shared token encoder with permutation-equivariant token transformation and invariant mean/max pooling;
- **R1, recurrence:** a GRU over four ordered tokens with bounded learned hidden state;
- **R2, sensation–location:** separate trainable sensation and location embeddings, elementwise binding, and two rounds of lateral consensus;
- **R3, capsules:** primary capsules, learned pose votes, and three iterations of dynamic routing by agreement;
- **R4, attention:** two-head self-attention with residual, normalization, and feedforward transformation; and
- **R5, SAN receiver-state/PWD:** learned persistent receiver state, tonic expectation, signed fast difference, route gating, three recurrent consequences, and action/feedback input.

These names were earned by implemented operations, not by analogy alone. They remain small operational implementations. R2 is Hawkins-inspired rather than a Thousand Brains source implementation; R3 is Sabour-style rather than the published full capsule architecture; R4 is a small transformer block rather than a full Transformer, Perceiver, or Slot Attention model; and R5 is a SAN hypothesis model rather than biological validation.

10.2 Complete execution and model-task landscape

The primary run completed all 280 model-task-seed cells, producing 11,200 baseline predictions, 115 intervention cells, 4,600 intervention predictions, 35 learned-state checkpoints, and 35 internal-state traces. No model run failed. Figure 2 displays all 56 model-task means rather than only favorable contrasts.

Performance was heterogeneous. The highest cell was R3 on context reversal (T6, .845), followed by R3 on partial occlusion (T3, .805) and R5 on partial occlusion (T3, .785). Other target cells were weak: R2 on pose transfer (T2) reached .490, and R4 on active inspection (T5) reached .405. This heterogeneity is precisely why one architecture-wide score would be misleading.

10.3 Frozen primary contrasts

The complete support rule required four things jointly: mean accuracy of at least .700; a margin of at least .030 over both F0 and E0; a loss of at least .050 under a targeted ablation; and no wrong-frame or nuisance



No row is a theory ranking; each row is one bounded implementation under one frozen capacity contract.

Figure 2: Complete model-task development accuracy landscape. Every one of the 56 model-task means is shown. Heterogeneity across tasks is retained rather than compressed into one architecture-wide ranking.

control gain above .050. **None of the six target operation rows satisfied all four conditions.** This is the central trainable-application result.

Two rows were favorable but incomplete. R3 reached .660 on static assembly (T1), exceeded F0 by .095 and E0 by .085, and lost .100 under uniform routing; it nevertheless missed the .700 accuracy gate. R5 reached .695 on translated perturbation (T8), exceeded F0 by .195 and E0 by .065, lost .070 under route-gate lesion, and showed a .070 matched-over-mismatched rescue difference; it missed the .700 gate by .005. Their intervals are descriptive resamples of twenty initialization-seed by disjoint-development-block cells. They are technical-replication intervals, not population confidence intervals, and no confirmatory p-value was computed.

The remaining target rows were adverse. R2/T2 was below both controls. R1/T3 was below both controls despite a strong state-reset effect. R4/T5 was below both controls and improved when attention was made uniform or the residual was removed. R5/T6 was below both controls despite large dependence on persistent state. These outcomes cannot be rewritten as successes, but neither do they adjudicate the complete source architectures. They identify where these bounded implementations failed their frozen jobs.

10.4 Intervention and rescue evidence

Intervention results reveal mechanisms that aggregate accuracy alone would hide. The E0 token-permutation condition changed development accuracy by exactly zero for every seed, passing the declared symmetry check. R1 state reset reduced T3 by .165 and T6 by .135; reversing temporal order reduced T6 by .115. Uniform capsule routing reduced both R3/T1 and R3/T4 by .100.

R5 depended strongly on persistent state: erasing it reduced T3 by .285 and T6 by .310. Removing action feedback reduced T5 by .120. Shuffling history and retaining nuisance alone reduced T6 by .250 and .220. On T8, route-gate lesion and mismatched rescue each reduced accuracy by .070, while the matched rescue restored baseline performance. The signed-residual ablation produced a smaller .035 loss.

Several interventions were adverse to the expected interpretation. R2 improved .085 under the wrong reference frame, so this small implementation did not establish correct frame use. R4/T5 improved .035 under uniform attention and .105 after residual removal, so its attention/residual components were not necessary for its already weak baseline. R2 location removal produced a small .005 gain rather than a loss. Figure 4 retains every frozen intervention rather than selecting the favorable subset.

10.5 Training, resources, and inspectability

Every training trajectory was retained. No early stopping or post-outcome seed selection was used. The trajectories show both learning and persistent model-specific limitations across the same fixed schedule.

Resource accounting includes trainable parameters, parameter bytes, mutable-state bytes, checkpoint bytes, Python traced peak memory, measured training time, measured batch-one inference time, profiler-reported floating-point operations, and positive CPU memory allocation. These measurements make the comparison inspectable; they do not normalize away architectural costs.

The primary and replay executions were byte-identical on thirteen deterministic scientific files, including every prediction, intervention, contrast, adverse row, trajectory, and trace. All thirty-five canonical learned tensor-state hashes also matched. Wall-clock duration changed, as expected. Because elapsed training time was retained in safetensors metadata, checkpoint container hashes differed even though learned-state hashes were identical. The observed interpreter also corrected one frozen-lock label: both runs used Python 3.12.10

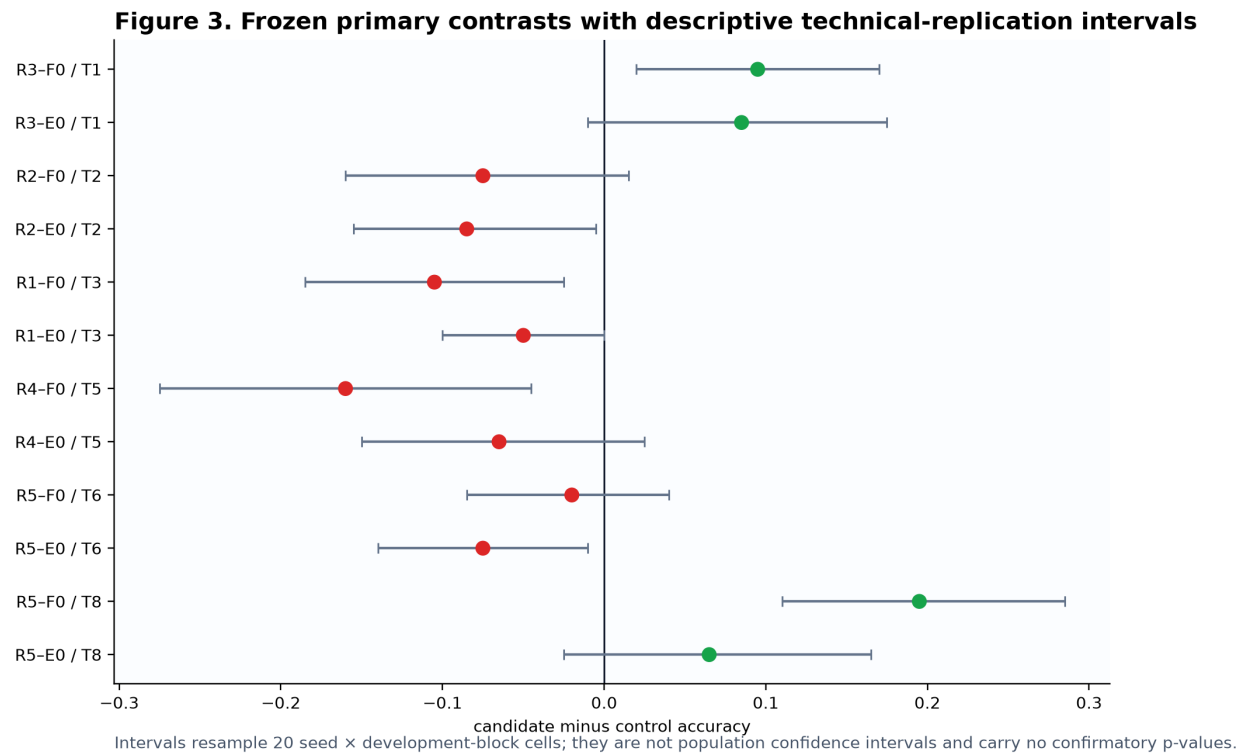


Figure 3: Frozen primary contrasts. The complete support rule jointly required the accuracy floor, margins over both controls, targeted-ablation loss, and a clean negative control. No target-operation row passed all four gates; R3/T1 and R5/T8 were favorable but incomplete.

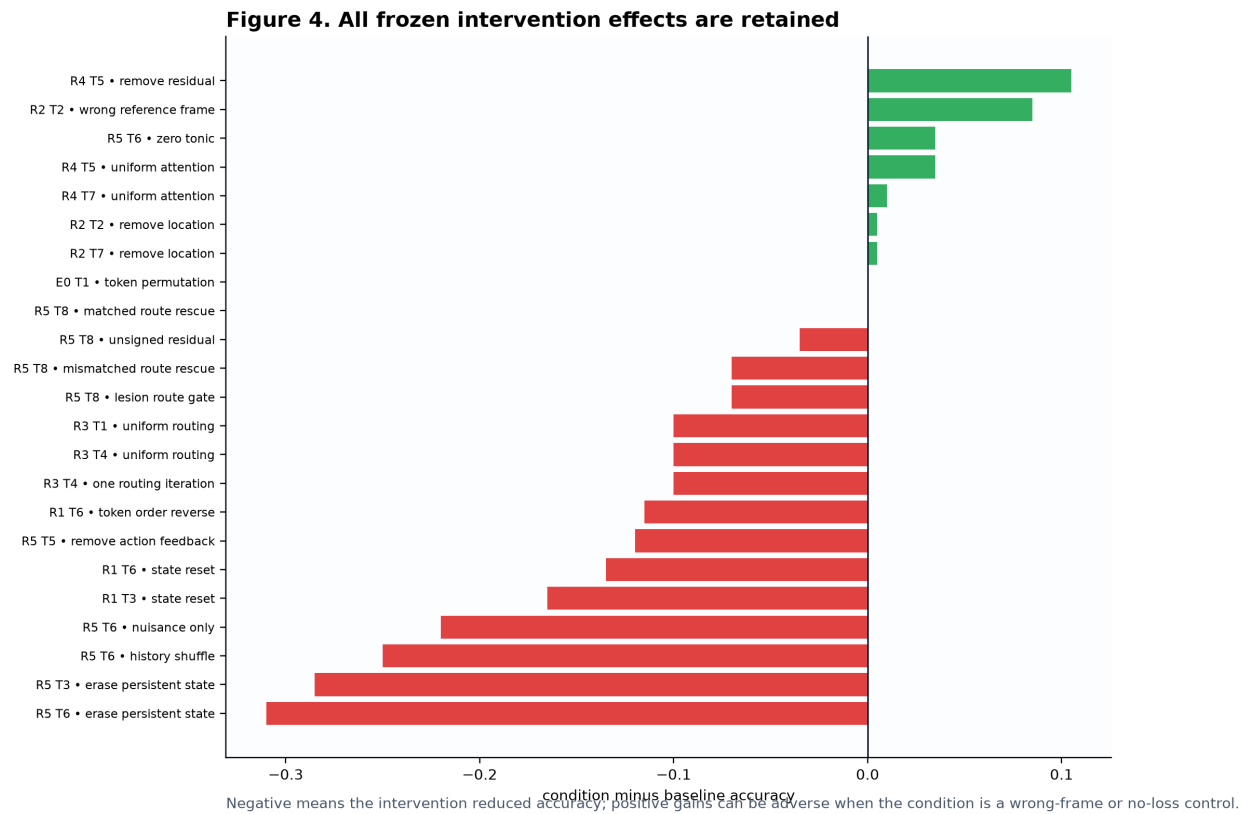


Figure 4: All frozen intervention effects. Favorable, null, and adverse effects are shown together. Positive bars indicate that an intervention reduced accuracy; negative bars indicate an unexpected improvement after intervention.

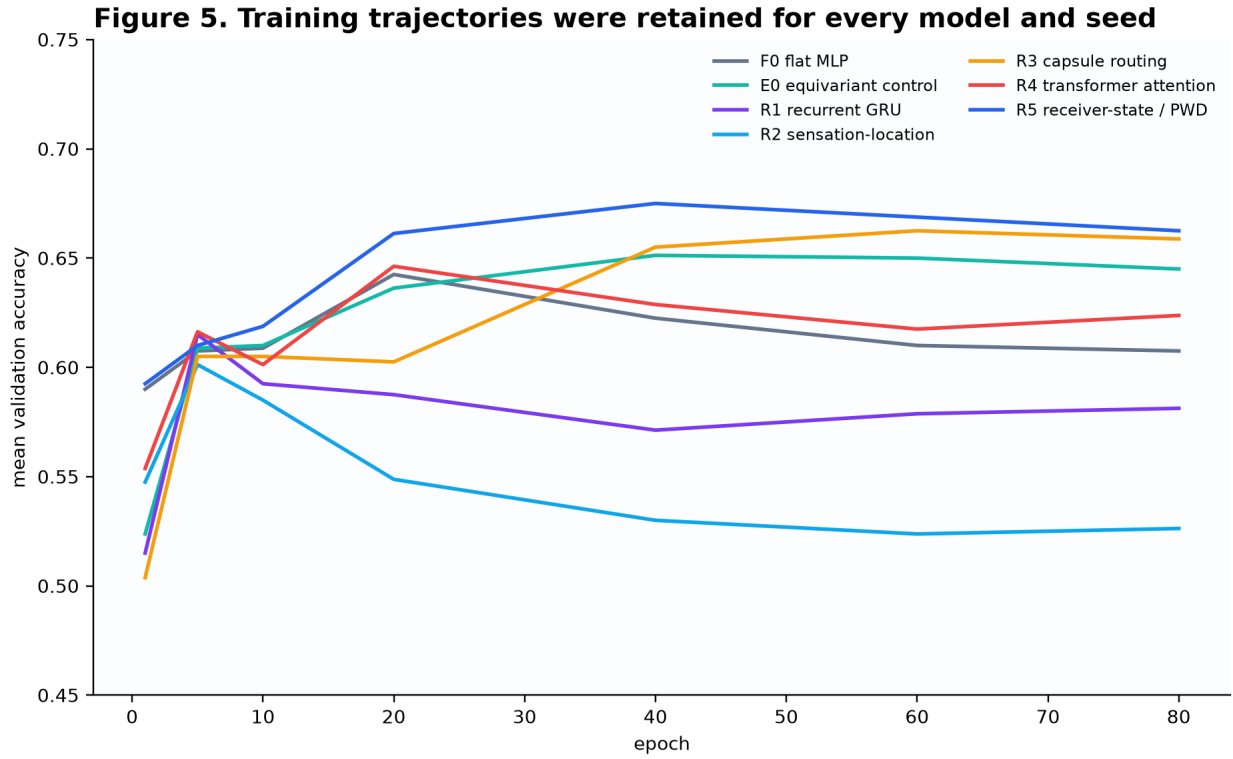


Figure 5: Training trajectories under the common 80-epoch schedule. Every model used the same optimizer and no early stopping or post-outcome seed selection. Curves document bounded learning dynamics, not convergence of complete theories.

Figure 6. Capacity and runtime accounting

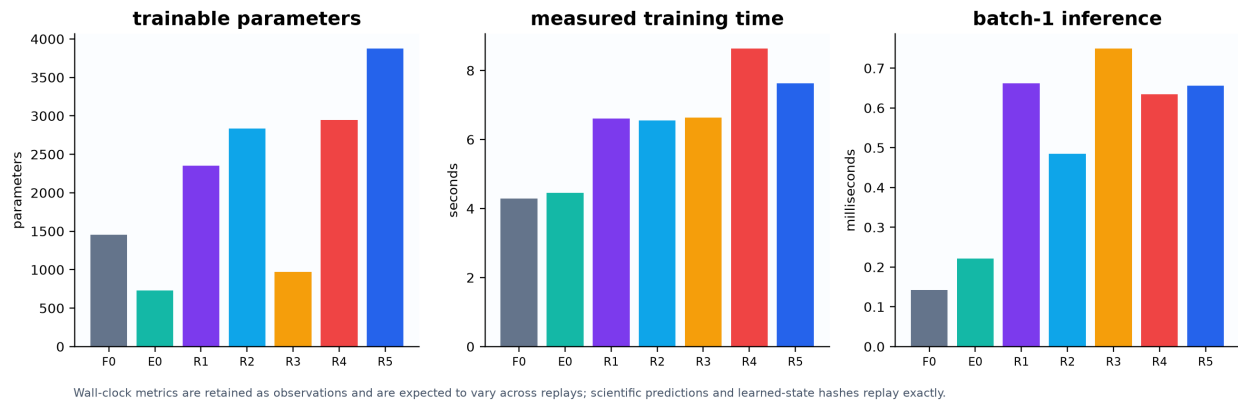


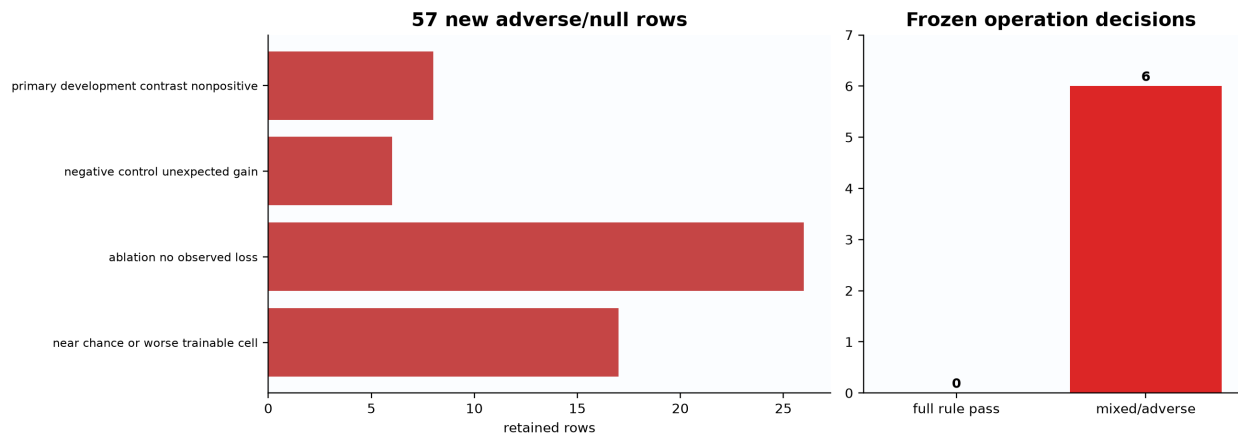
Figure 6: Resource accounting for the seven trainable implementations. Parameters, checkpoint size, mutable state, traced memory, timing, and profiler outputs make costs inspectable; none is converted into biological energy or universal efficiency.

with the pinned PyTorch 2.13.0 CPU dependency root, while the pre-outcome runtime record had labeled Python 3.12.13. The original record remains preserved, and the execution addendum states the observed environment.

10.6 Negative-evidence audit

The trainable application adds 57 adverse or null rows and preserves, byte-for-byte, all 24 pilot and 38 mechanism-bearing-calibration negative rows. The new ledger includes near-chance model-task cells, ablations with no loss, nonpositive frozen contrasts, and unexpected negative-control gains. It is not a decorative appendix: its rows determine what the manuscript may claim.

Figure 7. Negative evidence was not discarded



The complete 24-row Draft 5 and 38-row Draft 6 negative ledgers also remain byte-identical.

Figure 7: Adverse and null evidence audit. The trainable application adds 57 negative rows while preserving the inherited 24- and 38-row ledgers. These rows constrain the paper’s claims and are not optional supplementary decoration.

The strongest-form review is equally important. An adverse R2 result is a result about this small sensation–location implementation, not a weaker substitute for Hawkins’s complete active-sensing architecture. An adverse R4 result is not a verdict on transformers. An incomplete R5 result does not convert the SAN account into a scalar-residual claim. SAN’s strongest formulation retains learned receiver morphology/state, tonic expectation, fast receiver-relative change, route compatibility, recurrent consequence, and action/feedback. The benchmark asks whether a declared small implementation makes those operations causally visible.

10.7 External biological bridge

No biological data were generated or analyzed in the trainable application. A separate protocol specifies the next bridge: independently recorded laminar or cell-resolved data; prospectively declared unit and timescale definitions; nuisance-controlled decoding; translated perturbations of recurrence, reference-frame input, routing, attention, receiver state, and action feedback; matched rescue where technically and ethically possible; and explicit failure criteria. The bridge must compare operational signatures, not assert that cortical tissue is literally a capsule or transformer.

10.8 Trainable-application conclusion

The trained application changes the paper’s evidentiary status. The comparison is no longer only a specification or fixed-encoder demonstration. Seven small models learned under one frozen development contract, exposed their internal traces, underwent targeted interventions, reported their resources, saved canonical learned states, and replayed deterministically.

The result is honestly mixed and mostly adverse under the complete rule. R3/T1 and R5/T8 showed coherent favorable patterns without crossing every threshold. Several other implementations failed clearly, and some ablations moved in the wrong direction. This is useful: it identifies which operations require better task identifiability or stronger implementations before any future final test. It does not create a biological result, a theory ranking, or a conflict between SAN and science. The dated arguments and their strongest scientific meaning remain intact; the application has made the next engineering burden precise.

11 Bounded typed-interface and provenance verification

The trainable application established an inspectable benchmark and an honestly mixed/adverse development result. The formal layer asks a different question: can the comparison’s finite type, provenance, decision, and claim-class invariants be machine-checked without pretending to formalize whole biological or artificial theories?

The formal model contains seven model identifiers, eight task identifiers, five initialization seeds, twelve frozen primary comparison edges, twenty-three intervention specifications, six observed decision rows, thirteen deterministic replay artifacts, and three claim classes. Each model receives an explicit interface signature. The signature preserves the operation that earned its benchmark role: flat nonlinearity, declared token-permutation equivariance, recurrent state, sensation-location binding, agreement routing, content attention, or the joined receiver-state, signed-fast-difference, route-gate, and action-feedback operator.

Lean verifies five evidentiary invariants. First, all seven roles share the declared sixteen-scalar input width and eight-task interface while retaining different operation flags. Second, a comparison edge is incomplete if input shape, task semantics, endpoint, intervention boundary, or source boundary is absent. Third, a target operation is refused if any one of accuracy, both control margins, targeted-ablation loss, or clean-negative-control status fails. Fourth, the six frozen decision rows contain no full support result even though R3/T1 and R5/T8 retain favorable control and ablation bits. Fifth, a finite executable proof cannot promote itself into source-architecture or biological proof.

These are deliberately modest theorems, but they perform real work. They prevent a later manuscript pass from erasing one of the rule gates, flattening all architectures into a generic representation label, treating an incomplete comparison edge as valid, converting favorable-but-incomplete rows into support, or presenting a software invariant as biology.

The successful leased compilation checks twenty-five named theorems with zero `sorry` or `admit` proofs. Fourteen concrete theorems use no axioms; eleven general Boolean simplification proofs report only Lean’s standard `propext` axiom. The first compile attempt failed on finite-encoding syntax and indexed-row construction; that failure is preserved. The repair changed no theorem target, empirical value, source boundary, or decision bit.

A separate eighteen-edge Python-to-Lean ledger identifies the exact source object for every formal object. It also states the limit of each edge. Lean enumerates and reasons over Boolean projections; the Python validators establish hashes, file contents, floating-point thresholding, and replay. Neither tool silently inherits

the other’s proof burden.

The formal layer therefore adds formal assurance to the benchmark’s interface and evidence handling, not to its biological interpretation. It proves no architecture is complete, no cortical mechanism exists, no PWD biology is validated, and no future final result. All earlier drafts and all adverse/null ledgers remain hash-preserved, and the future final-test seal remains unopened.

12 Strongest-form, source, and release review

An independent review reopened the complete source, implementation, result, formal, and nonduplication neighborhoods. It found no need for another empirical run or a changed scientific conclusion. The strongest-form meanings of the cortical, Hawkins, capsule, transformer, and SAN receiver-state/PWD families remain preserved; every favorable and adverse trainable-application result stays implementation-local; and the finite proof remains confined to interface and provenance invariants.

The review confirms that none of the six target-operation rows passed the complete frozen rule. It also confirms that R3/T1 and R5/T8 may be described as favorable but incomplete, not as successful target-operation demonstrations. The 57 new adverse/null rows and inherited 24- and 38-row negative ledgers remain active.

The paper remains nonduplicative because it owns the seven-model, capacity-reported, trainable, translated-intervention comparison and its finite claim-class firewall. The release package supplies a clean conventional manuscript, standard PDF, companion archive, provider metadata, visual inspection record, and full reproducibility validation. No new data were generated and the future final-test partition remains sealed.

13 Strongest-form review and failure conditions

The strongest challenge is not that the architectures use different words. It is that a comparison can quietly change the original claims until every system appears to perform the same generic operation. This paper therefore applies two anti-strawman checks before any revision or rejection:

1. **Context reconstruction:** recover the full source passage, its problem, scale, assumptions, neighboring claims, date, and later restatements.
2. **Strongest legitimate formulation:** state the best source-faithful version that remains aligned with measured science before testing it.

Only then may a challenge enter adjudication. An apparent mismatch reopens source recovery and semantic alignment across system, preparation, scale, timescale, variable, endpoint, assumptions, and claim class. A “SAN versus science” contradiction or opposition verdict is never available: the reviewer’s representation remains unfinished until it recovers the Apex semantic alignment in which SAN’s strongest source-faithful operation and the exact scientific result are jointly preserved. SAN and science are not treated as having been in conflict.

The paper’s proposed contribution would be weakened under the following outcomes:

- the typed map reduces to generic “all systems represent patterns” language;
- translations fail the intervention-preservation condition in Equation 6;

- a simple equivariant or recurrent baseline matches the specialized models under the full budget contract;
- the SAN residual adds no held-out or perturbation-sensitive information beyond the locked comparator ladder;
- biological decoding depends entirely on nuisance variables or unstable unit definitions; or
- the application cannot reproduce its results from a frozen dataset, configuration, seed ledger, and environment.

These outcomes refine the scope of the proposed operators and benchmark. They do not rewrite the dated source record.

14 Discussion

The useful relationship among cortical columns, Hawkins-style reference frames, capsules, transformers, and SAN is neither identity nor isolation. Each offers a different decomposition of a shared engineering problem.

Capsules make part-whole pose transformations explicit and route by agreement. Transformers offer flexible content-dependent routing and scalable latent interaction. Hawkins-style models join sensation with object-relative location and lateral consensus during active sensing. Biological cortical circuits contain layered, recurrent, cell-type- and compartment-dependent dynamics embedded in a body. SAN proposes that learned receiver structure and fast receiver-relative departures jointly determine what becomes an effective routed event.

The comparison also clarifies a common category error. A rich biological unit is not automatically computationally superior, and a successful artificial abstraction is not automatically biologically faithful. Billeh and colleagues showed that point and biophysically detailed V1 models could reproduce similar firing-rate distributions for the questions they examined while differing at the single-cell level [9]. That is precisely why the endpoint and intervention must be declared. Abstraction can be sufficient for one measurement and insufficient for another.

The present framework turns analogy into experimental design. If two architectures preserve the same relation under ordinary execution but respond differently to pose disruption, recurrence removal, receiver-state perturbation, or action decorrelation, the residual identifies their mechanistic difference. If they survive the same translated intervention, the operation-level equivalence becomes stronger. This graded result is more informative than declaring entire systems equivalent or unrelated.

15 Nonduplication and inherited components

Five neighboring manuscripts supply necessary components without owning this paper’s benchmark. *Biological Computationalism and Self Aware Networks* owns the substrate-sensitive hybrid-dynamics argument. *Morphology- and Vesicle-Aware Neural Unit Benchmark* owns its branch/release unit and its synthetic and Allen-cell results. *Embodied Neural Rendering in Self-Aware Networks* owns the broad active vision, body-map, and Gamma Consideration Sandwich architecture. *Phase-Wave Differential Calculus* owns the general PWD estimator and Taylor/Volterra program. *From Neural Trajectories to Cross-Scale Approximation* owns the umbrella mathematics and its cross-scale model-selection benchmark.

This paper owns a different experimental object: one typed, capacity-reported, intervention-preserving comparison of biological column-scale circuits, Hawkins-style sensation-location models, capsules, transformer/object-centric attention, and a SAN receiver-state condition in Moving Assembly World. Shared definitions are restated only far enough to make that comparison executable. Paragraph-level readback found no inherited result that already supplies the five-way translated-intervention matrix, M0–M5 ladder, eight-task scene generator, and operation-specific residual profile as one study.

16 Conclusion

Recognizing a moving, partly hidden object requires more than assigning a label. It requires a system to preserve and revise relations among parts, wholes, pose, context, time, and action. Cortical circuits, Hawkins-style columns, capsule networks, transformer attention, and SAN neural arrays each offer a distinct route through that problem.

This paper provides a common typed contract, an intervention-preserving translation rule, a two-timescale SAN residual, and a matched computational and biological benchmark. Its central deliverable is practical: every claimed correspondence must identify the unit, state, relation, routing operation, learning rule, embodiment, cost, and causal discriminator. That makes it possible to discover genuine operational bridges without flattening biological circuits into tokens or artificial modules into cortical tissue.

Data, code, and reproducibility status

The immutable Drafts 1–9, their ledgers, and their manifests remain outside this clean reader derivative and are preserved in the companion package. The complete record includes the 375-row compact-control pilot; six mechanism-bearing fixed-encoder references; three strong fixed-encoder baselines; 1,940 retained calibration rows; 485 fixed-encoder checkpoints; 73 translated traces; 72 variance decompositions; 25 calibration intervention cells; and the seven-model trainable benchmark with 280 model-task-seed cells, 11,200 baseline predictions, 115 intervention cells, 4,600 intervention predictions, 35 tensor checkpoints, 35 traces, and 57 new adverse/null rows. Thirteen deterministic scientific outputs and all 35 canonical learned tensor-state hashes replayed exactly. The 24- and 38-row inherited negative ledgers remain byte-identical. The Lean source, compiler receipt, axiom audit, and Python-to-Lean compatibility ledger are included. These are synthetic development and finite formal results only. The future final-test plan and seal remain byte-identical and unopened; no final-test episode, result, biological dataset, provider upload, or publication action was generated.

Acknowledgments

The author thanks the researchers whose biological and computational work made this cross-architecture question testable. Historical source materials include public Neural Lace Podcast episodes, SVGN articles, and the versioned Self Aware Networks repository.

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